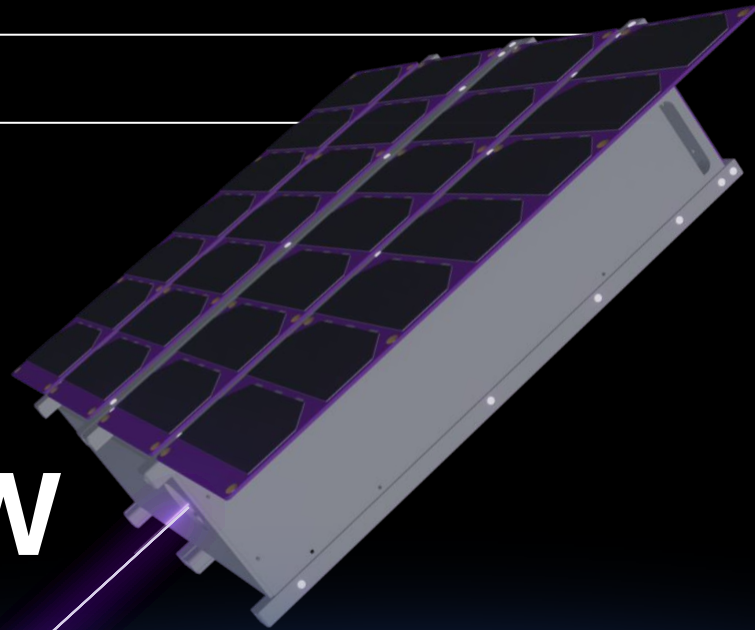

UCHICAGO SPACE PROGRAM

PULSE-Q MISSION CONCEPT REVIEW

JANUARY 23 2026



Polarization-Modulated Laser Satellite Experiment – Quantum

PULSE-Q

Mission Concept Review

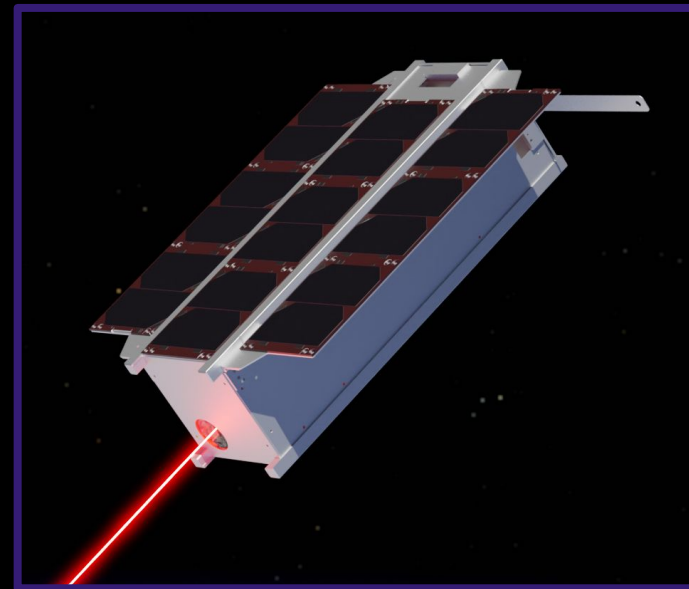
Introduction

UCSP Development Team Introductions

Logan Hanssler	Project Director	Ellisian King	Optical Engineer
Stew Kristiansen	Chief Engineer	Eric Lin	Mechanical Engineer
Ella Gentry	Systems Engineer	Lexi Liu	Mechanical Engineer
Nelson Chan	Quantum Engineer	Phil Qi	Mechanical Engineer
Neha Chandran	Quantum Engineer	Mika Sadikario	Electrical Engineer
Unnathi Venkatesh	Quantum Engineer	Sheel Pant	Software Engineer
Selina Gao	Optical Engineer		

PULSE Program Overview

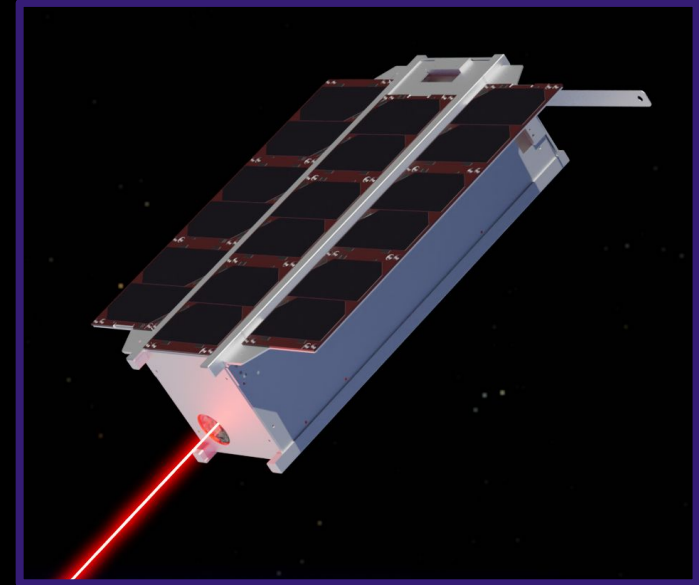
- ❑ PULSE-A is a ≤ 6 U CubeSat mission aiming to conduct space-to-ground optical communications using circular polarization modulation
- ❑ PULSE-Q is a follow-up 6U CubeSat mission aiming to conduct space-to-ground quantum key distribution (QKD) over an optical transmission
 - ❑ PULSE-Q builds upon the institutional knowledge, designs, hardware, and software created for PULSE-A
- ❑ Both development teams are led by undergraduate students in the UChicago Space Program and are primarily driven by undergraduate design work



PULSE-A CubeSat render as of 11/23/2025

PULSE Program Overview

- ❑ PULSE-A and PULSE-Q each feature a custom payload transmitter, optical ground station (OGS) receiver, and spacecraft bus
 - ❑ Both missions will utilize the same RF ground station (RFGS) during mission operations
- ❑ The PULSE program was founded in February 2023 beginning with concept studies for a space-to-ground QKD mission
 - ❑ Development shifted to PULSE-A as a risk reduction mission in August 2023
 - ❑ Dedicated development for PULSE-Q has restarted as of September 2025



PULSE-A CubeSat render as of 11/22/2025

Mission Concept Review (MCR) Purpose

- ❑ The MCR affirms the mission need, examines the proposed mission objectives, and outlines the early stage concept for meeting those objectives
- ❑ The MCR examines the feasibility of accomplishing the mission objectives within the proposed schedule and budget
- ❑ The MCR previews and finalizes areas of focus for the upcoming system concept development process
- ❑ The MCR provides first-round merit and feasibility feedback in advance of the mission's proposal to the 2026 Mission Concept RFP, issued by the University Nanosatellite Program (UNP)

MCR Agenda

1. Mission Overview
2. Schedule, Budget, & Funding
3. Optical System Concept: Quantum Overview
4. 10-Minute Intermission
5. Optical System Concept: Optical Overview, Data Processing
6. Spacecraft System Concept
7. Risk Assessment & Mitigation
8. Conclusion

Not Included in the MCR

- ❑ System requirements beyond the Principal Mission Objective
- ❑ Fully developed concepts for system and subsystem architectures
- ❑ Component selections
- ❑ Component-level mass, volume, and financial budgets

Requested Feedback Format

- ❑ Reviewers are encouraged to interject the MCR presentation at any time with questions or comments
- ❑ Please note that we may request to take individual discussions offline in the interest of keeping the review on schedule
- ❑ The MCR Reviewer Response Sheet has been provided to all reviewers over email
- ❑ While we would prefer for written feedback to be submitted using the sheet template, we sincerely appreciate feedback provided in any format most convenient for the reviewer
- ❑ We request all written feedback be submitted by January 30th, 2026 and NLT February 6th, 2026

The image shows a stack of 'PULSE-Q REVIEWER RESPONSE SHEET' forms. The top form is titled 'PULSE-Q Polarization-modulated Laser Satellite Experiment - Quantum' and includes sections for 'Introduction', 'Mission Overview', 'Schedule, Budget, & Funding', and 'Optical System Concept: Quantum Overview'. The forms are white with purple borders and text.

Mission Overview

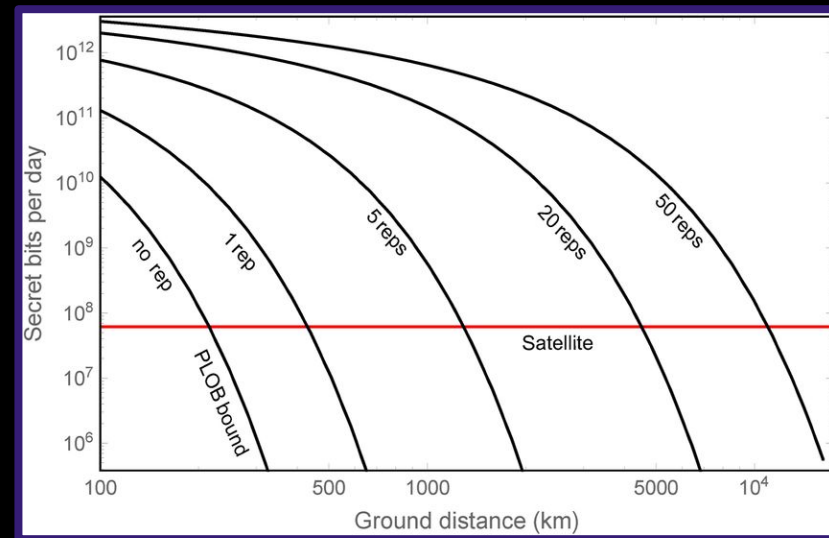
Principal Mission Objective

The mission shall execute satellite-to-ground data transmission using single-photon level laser communications, qualified by:

- ❑ QKD with 128 bits (TBR) $\leq$ Final Key Length $\leq$ 2,048 bits (TBR)
- ❑ $2 \leq$ Number of Polarization States $\leq$ 4 for encoding data
- ❑ 5 seconds $\leq$ Consecutive Transmission Time $\leq$ 150 seconds
- ❑ A transmission terminal with Volume $\leq$ 2U and Cost $\leq$ \$110,000 (TBR)
- ❑ An open-source architecture (pending intellectual property disclosure policies)

Mission Merit

- ❑ QKD stands on the threshold of becoming a viable method for secure global communications
- ❑ Ground-based fiber networks suffer from distance limitations, especially considering major challenges in realizing quantum repeaters
- ❑ Provably reliable space-based QKD would enable secure communication networks to be established over large distances
- ❑ Prioritizing small form factor terminals would allow the system architecture to be more readily proliferated into a robust network



Ground-based QKD performance compared to the ideal case using a satellite node
Source: Sidhu et al. (2021)

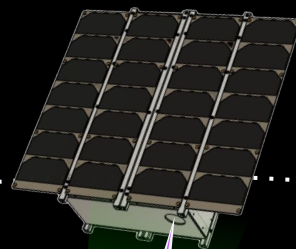
Mission Design Philosophy

- ❑ PULSE-Q serves a proof of concept for **commercializable, small form factor** single-photon optical communications terminals
- ❑ The mission is being developed with much stricter risk tolerance than the PULSE-A predecessor
 - ❑ Maintaining low SWaP+C and promoting undergraduate education remain significant mission priorities; however, these priorities are subordinate to minimizing mission risk
- ❑ PULSE-Q's development aims to repurpose as much of the design work from PULSE-A as possible
 - ❑ Many subsystems may need to be largely redeveloped to meet the differing mission requirements or purchased from commercial vendors to reduce the risk of failure
 - ❑ Institutional knowledge, documentation, partnerships, and program organization methods will be sustained throughout the PULSE Program

Mission Concept

Space Segment:

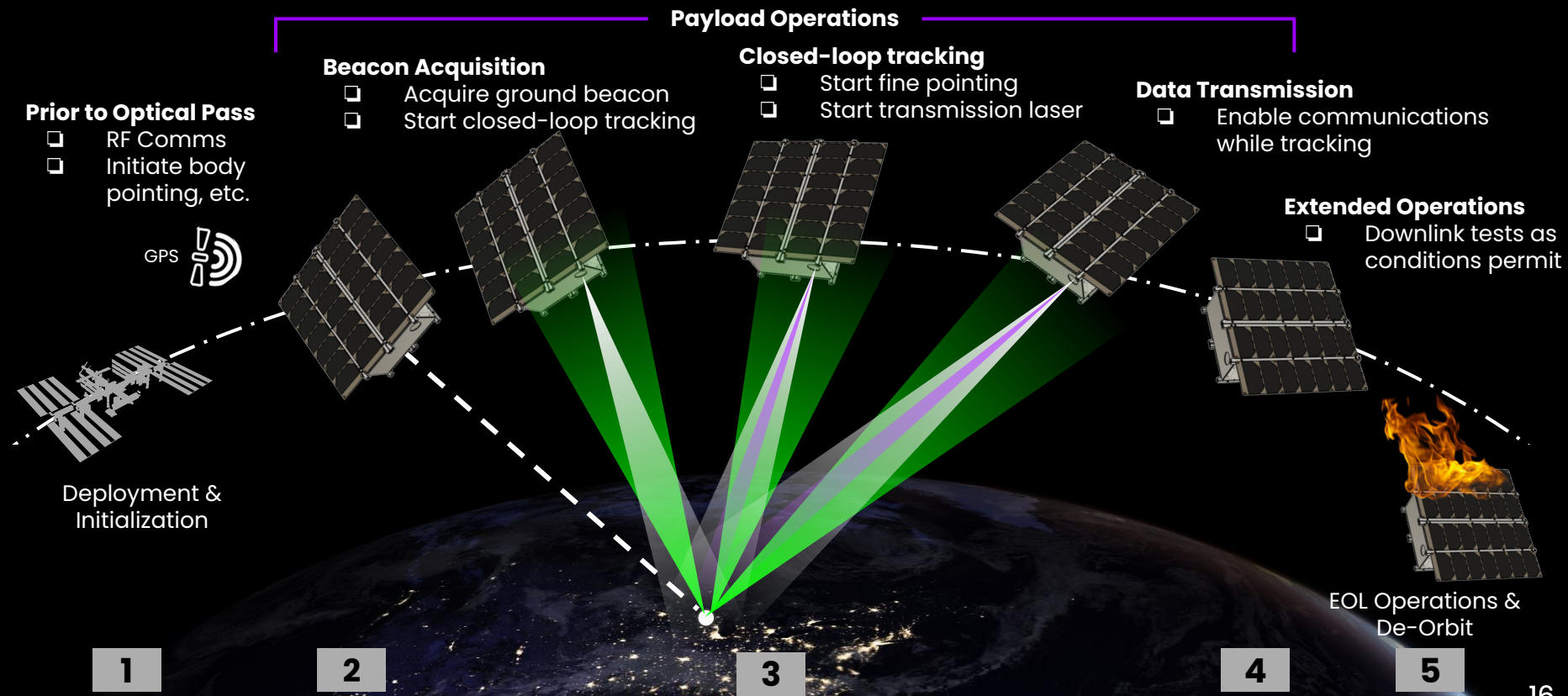
- ❑ Satellite transmits single-photon level optical downlink and payload beacon laser
- ❑ Satellite receives OGS beacon laser
- ❑ Satellite performs closed-loop tracking to maintain optical link
- ❑ Satellite performs optical instrument calibrations



Ground Segment

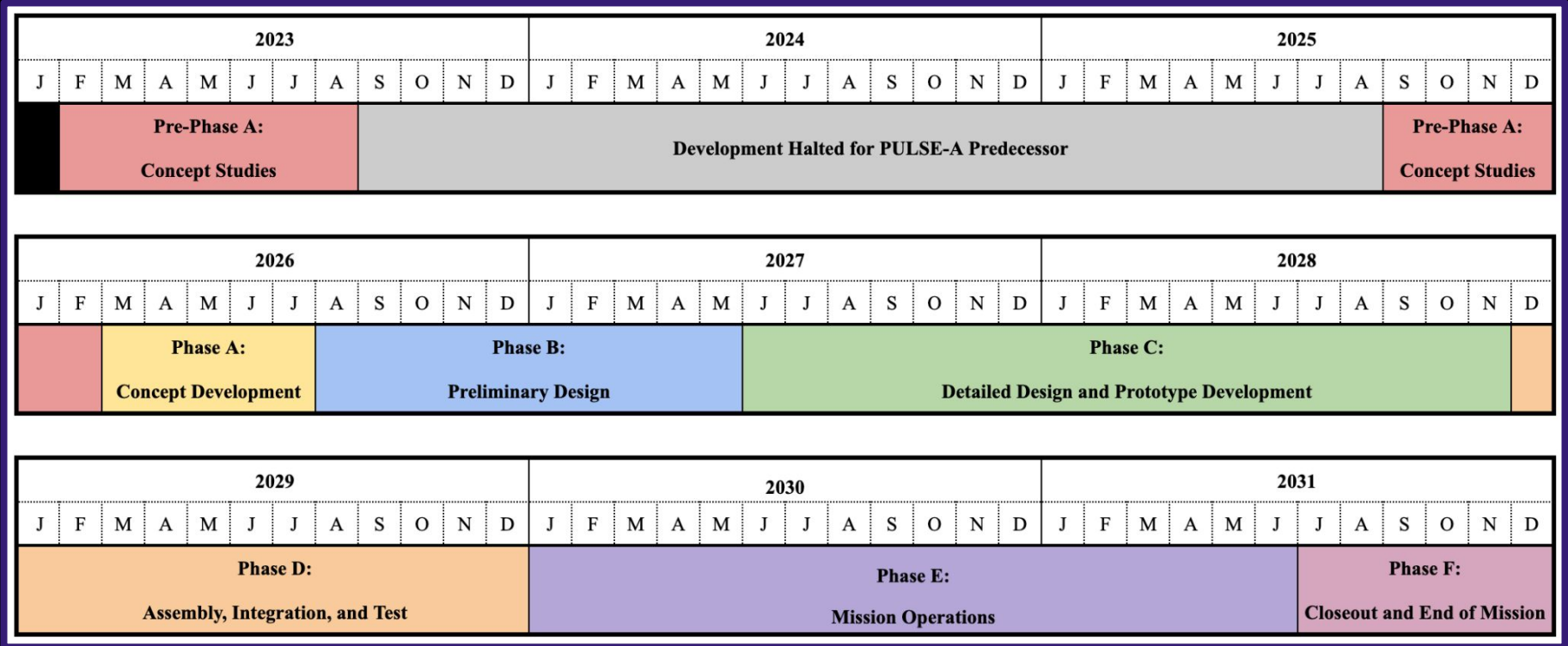
- ❑ OGS receives single-photon level optical downlink and payload beacon laser
- ❑ OGS decodes single-photon level signal to accomplish QKD
- ❑ OGS transmits beacon laser to satellite

Concept of Operations

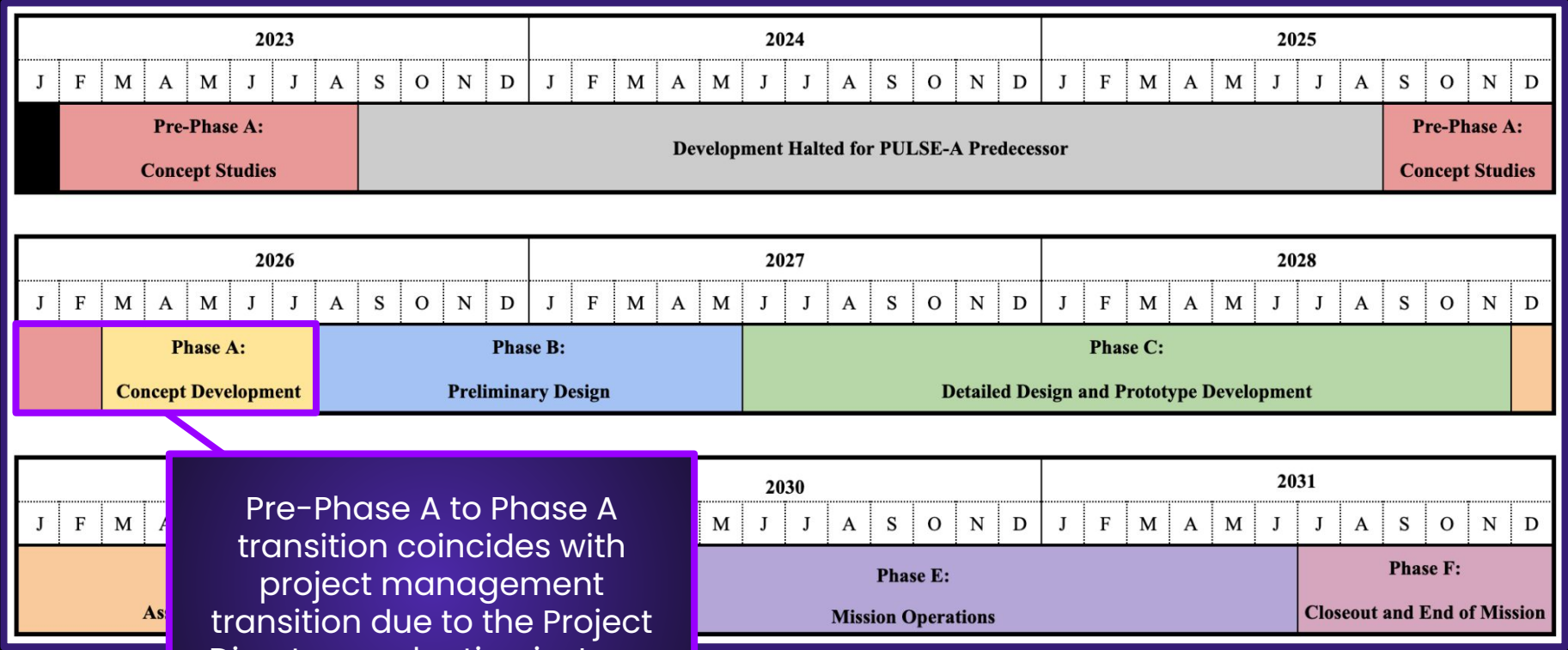


Schedule, Budget, & Funding

Development Timeline

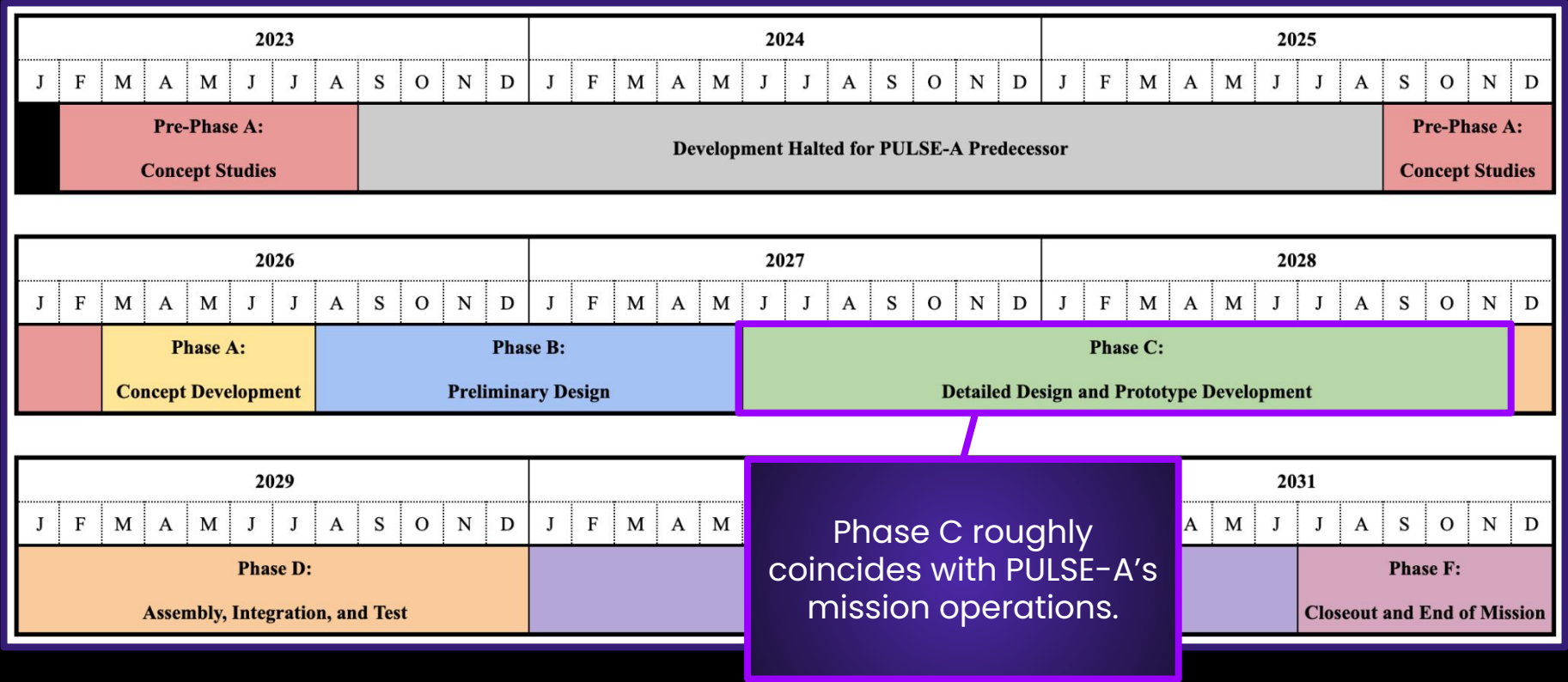


Development Timeline

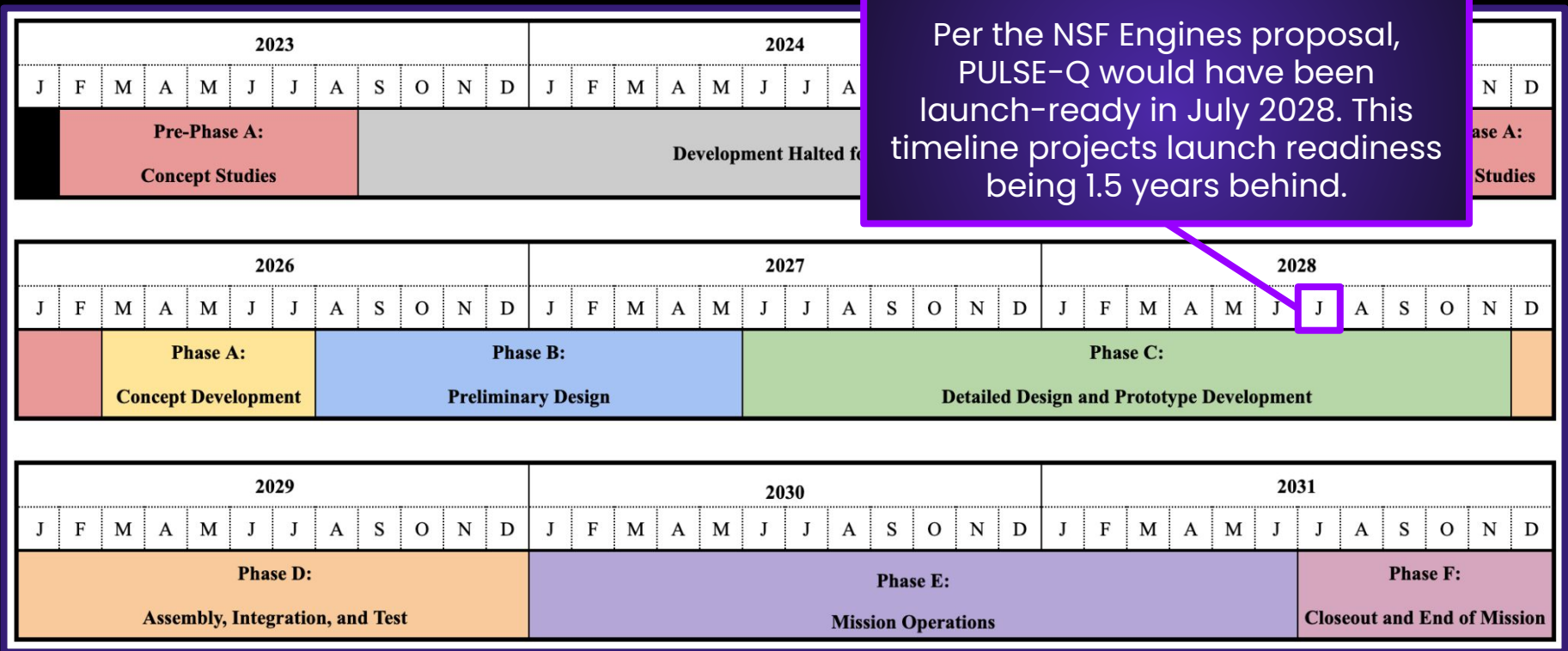


Pre-Phase A to Phase A transition coincides with project management transition due to the Project Director graduating in June.

Development Timeline



Development Timeline



Potential NSF Engines Funding Source

Budget Item	Funding Amount (\$)
Payload	110,000.00
Bus	200,000.00
Optical Ground Station	100,000.00
RF Ground Station	10,000.00
Assembly, Integration, & Test	10,000.00
Misc. Materials & Supplies	10,000.00
Personnel Salaries	200,000.00
Launch	0.00
Equipment Contingency	25,000.00
Total	665,000.00

Anticipated Development Budget

Budget Item	Budget Amount (\$)
Payload	110,000.00
Bus	200,000.00
Optical Ground Station	100,000.00
RF Ground Station	10,000.00
Assembly, Integration, & Test	40,000.00
Misc. Materials & Supplies	30,000.00
Personnel Salaries	200,000.00
Launch	400,000.00
Equipment Contingency	97,000.00
Total	1,187,000.00

Anticipated Development Budget

Budget Item	Budget Amount (\$)
Payload	110,000.00
Bus	200,000.00
Optical Ground Station	100,000.00
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Misc. Materials & Supplies	30,000.00
Personnel Salaries	200,000.00
Launch	400,000.00
Equipment Contingency	97,000.00
Total	1,187,000.00

Allows for ground twin of quantum/optical system to be developed and operated in Zhong Lab.

Anticipated Development Budget

Budget Item	Budget Amount (\$)
Payload	110,000.00
Bus	200,000.00
Optical Ground Station	100,000.00
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Misc. Materials & Supplies	30,000.00
Personnel Salaries	200,000.00
Launch	400,000.00
Equipment Contingency	97,000.00
Total	1,187,000.00

Allows for commercially-developed bus and a ground station telescope with a larger aperture than that of PULSE-A.

Anticipated Development Budget

Budget Item	Budget Amount (\$)
Payload	110,000.00
Bus	200,000.00
Optical Ground Station	100,000.00
RF Ground Station	10,000.00
Assembly, Integration, & Test	40,000.00
Misc. Materials & Supplies	30,000.00
Personnel Salaries	200,000.00
Launch	400,000.00
Equipment Contingency	97,000.00
Total	1,187,000.00

Small budget presumes the PULSE-A RFGS will be repurposed for PULSE-Q operations. This possibility is discussed on slides 84–85.

Anticipated Development Budget

Budget Item	Budget Amount (\$)
Payload	110,000.00
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Optical Ground Station	100,000.00
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Misc. Materials & Supplies	30,000.00
Personnel Salaries	200,000.00
Launch	400,000.00
Equipment Contingency	97,000.00
Total	1,187,000.00

Anticipated cost
for 6U CubeSat.

Anticipated Development Budget

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Optical Ground Station	100,000.00
RF Ground Station	10,000.00
Assembly, Integration, & Test	40,000.00
Misc. Materials & Supplies	30,000.00
Personnel Salaries	200,000.00
Launch	400,000.00
Equipment Contingency	97,000.00
Total	1,187,000.00

20% of above line items' sum, excluding launch.

Budgetary Gap

Budget Item	Gap Amount (\$)
Payload	0.00
Bus	0.00
Optical Ground Station	0.00
RF Ground Station	0.00
Assembly, Integration, & Test	30,000.00
Misc. Materials & Supplies	20,000.00
Personnel Salaries	0.00
Launch	400,000.00
Equipment Contingency	72,000.00
Total	522,000.00

Budgetary Gap

Budget Item	Gap Amount (\$)
Payload	0.00
Bus	0.00
Optical Ground Station	0.00
RF Ground Station	0.00
Assembly, Integration, & Test	30,000.00
Misc. Materials & Supplies	20,000.00
Personnel Salaries	0.00
Launch	400,000.00
Equipment Contingency	72,000.00
Total	522,000.00

Largest gap can be closed if a launch sponsorship is secured.

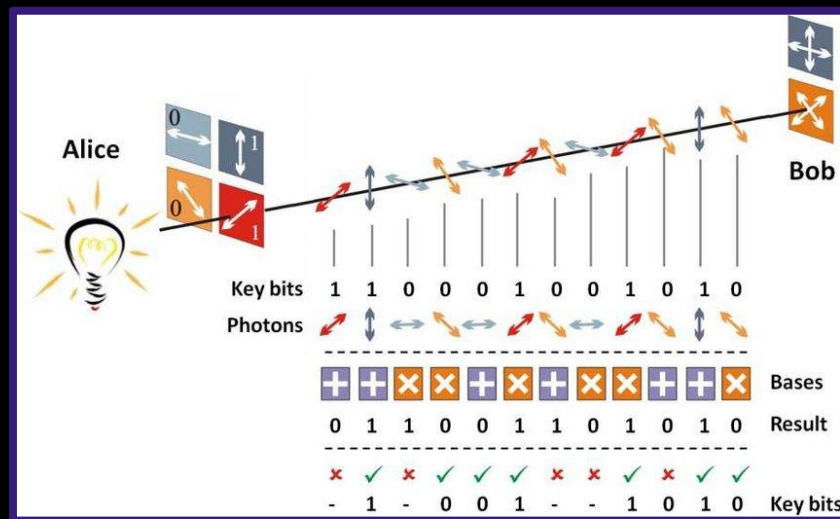
Closing the Budgetary Gap

- ❑ NSF Engines' success would fund nearly all non-launch costs
- ❑ University Nanosatellite Program may lead into a launch provider or additional sponsorships being secured
 - ❑ Other launch or component sponsors should be sought out and considered
- ❑ Donations may bring in up to \$100,000 over the project lifetime
 - ❑ PULSE-A is more likely to be the priority for any donations directed to UCSP
- ❑ We may consider selling bus space to a secondary payload provider, given the PULSE-Q CubeSat's anticipated ~2U volume margin

Optical System Concept Quantum Overview

BB84 Scheme

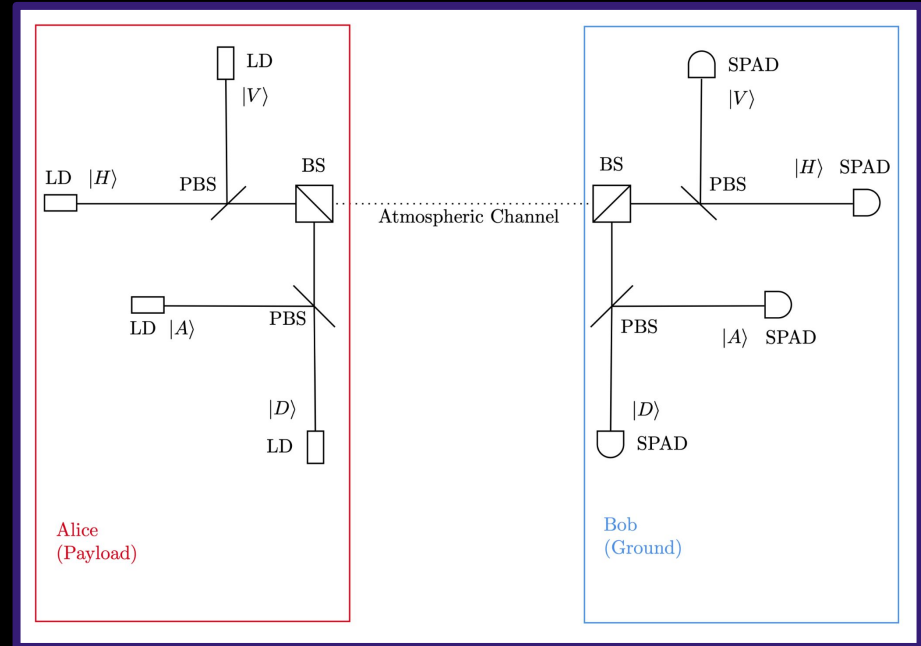
- Qubits give four separate qubit states together ("H/V"), ("D/A")
- Photons propagate to Bob
- Bob picks a random measurement basis and detects photons
- Alice and Bob compare bases and throw out mismatched events (sifting process)
- Errors are estimated and the remainder is the "quantum key"



Source: Carrasco-Casado et al. (2016)

BB84 Layout

- ❑ Four states modulated by polarization in the payload
- ❑ Four detectors on the ground
- ❑ Space-side is heavily space-constrained
- ❑ Ground-side is less space-constrained

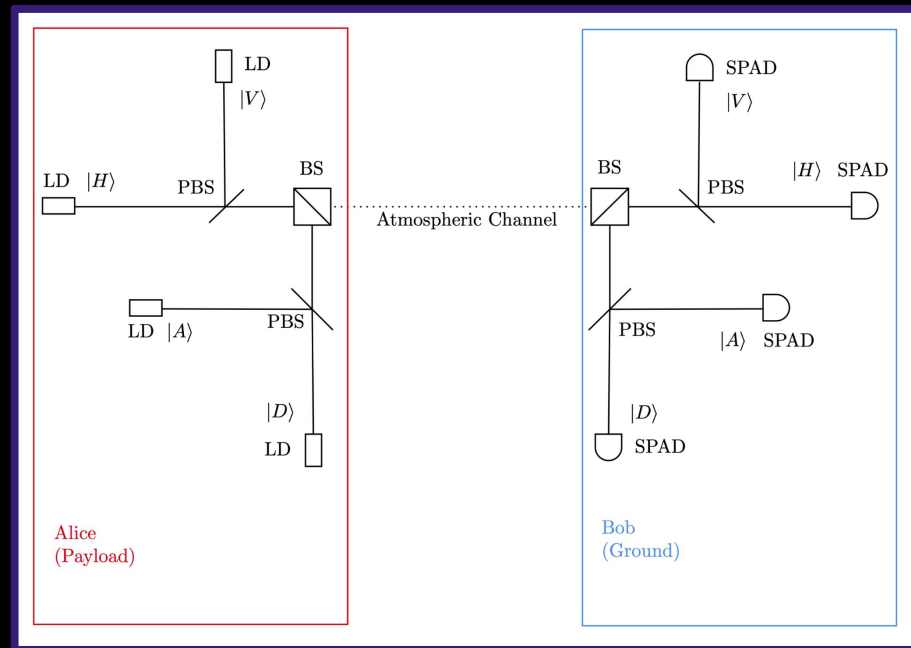


Credit: S. Kristiansen, PULSE-Q

BB84 Layout

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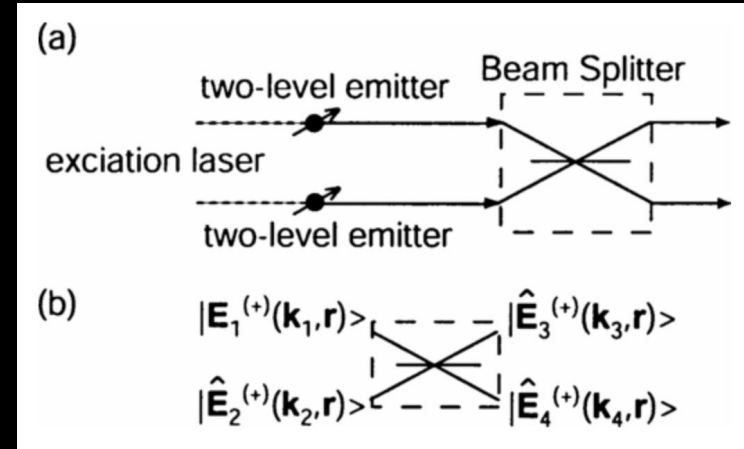
The use of four separate LDs may prove difficult due to SWaP constraints.



Credit: S. Kristiansen, PULSE-Q

Quantum Source

- ❑ The emitter is modeled as a **two-level system**, with ground state $|g\rangle$ and excited state $|e\rangle$.
- ❑ Optical/electronic excitation promotes the system to $|e\rangle$, followed by spontaneous radioactive decay that emits a **single photon**.
- ❑ Photon emission events are temporary anti-correlated leading to photon **antibunching**.



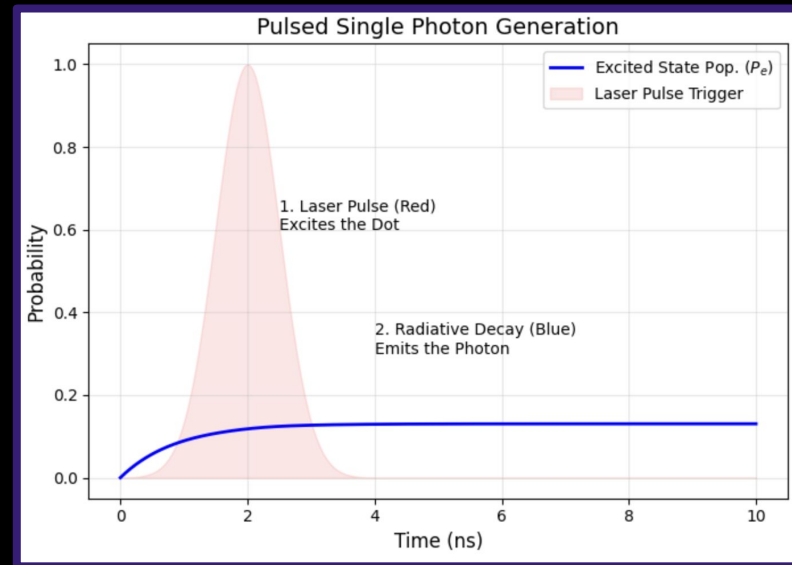
Source: Kiraz et al. (2004)

Sources of Quantum Losses

- ❑ Space Noise Sources
 - ❑ **Radioactive decay** caused by nuclear/cosmic radiation
 - ❑ **Spectral diffusion** caused by fluctuating electric charges
 - ❑ **Thermal dephasing** caused by temperature increase
 - ❑ **Stray light** from the sun's photons, Earth's albedo, or the moon
- ❑ Other Sources of Error
 - ❑ **Fluctuating temperature** (-65°C to 130°C) in LEO
 - ❑ **Pulsed excitation source** being used for QKD experiments as opposed to a continuous wave laser

Modeling Quantum Losses

- ❑ We can encode these parameters into “collapse operators” and feed them into the **Lindblad Master Equation** for open quantum systems.
- ❑ Using the **QuTip library in Python**, we can simulate time evolution of our quantum state.
- ❑ By extracting the number of **photons emitted per pulse**, we can determine whether our source exhibits quantum correlations.



Credit: N. Chandran, PULSE-Q

Modeling Quantum Losses

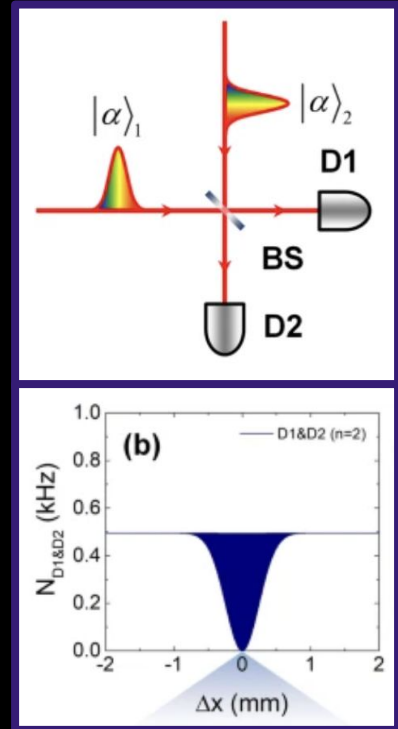
- ❑ Depending on the physical implementation, we can choose between simulating **continuous wave** (CW) or **pulsed excitation**
- ❑ In CW implementations, the Hamiltonian driving force will be a function of **Rabi frequency**, which is comparable to the decay rate
 - ❑ Then, we can calculate the **second-order correlation function, $g^2(\tau)$** , which characterizes photon antibunching and multi-photon emission probability
- ❑ In pulsed excitation implementations, the Hamiltonian is time-dependent and is a function of the amplitude of the laser's **Gaussian π -pulses**
 - ❑ The total number of photons is given by integrating the emission rate over time

Solutions to Quantum Losses

- ❑ **Cosmic / nuclear radiation:** Radiation-hardened detectors (SPADs/APDs) & shielding
- ❑ **Spectral diffusion:** Stabilized sources, wavelength filtering
- ❑ **Thermal dephasing:** Thermal control, polarization compensation:
- ❑ **Stray light:** Spectral & temporal filtering, site selection
- ❑ **Fluctuating temperature:** Thermal design, Active calibration
- ❑ **Pulsed excitation source:** High-stability pulsed sources, Time-tagging & gating

Quantum Interference

- ❑ Initial testing may need a **weak coherent pulse source** before obtaining a perfect single photon emitter
- ❑ Even with low photon count, phase randomized weak coherent pulses can display the same characteristic **Hong-Ou-Mandel (HOM) fringe** interference effect
- ❑ By **Dirac's principle**, which implies that individual photons do not need to be perfectly overlapped within an interferometer, our photons can still be adjusted to produce antibunching



Source: Kim et al. (2021)

10-Minute Intermission

Optical System Concept

Optical Overview, Data Processing

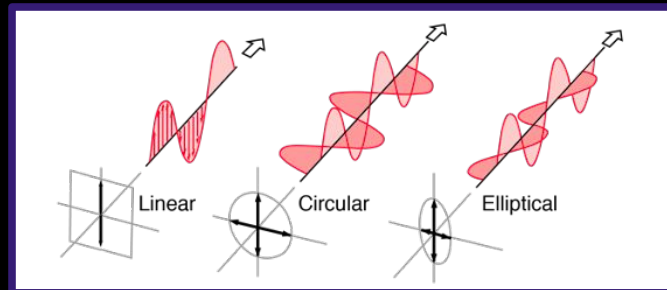
Basis Polarization State Selection

Polarization: 2 bases, 4 state

Presented as:

Rectilinear: (horizontal (H) / vertical (V))

Diagonal: (diagonal (D) 45° / antidiagonal (A) 135°)



Source: Nave, C. R. (n.d.). *Classification of polarization*. HyperPhysics.

Polarization States Chosen: **Circular (L/R) & Linear (H/V)**

We will not be using elliptical polarization as one of the two bases because it is more sensitive to phase shifts while offering no security advantage over linear polarization bases.

Preliminary Phase Modulator Trade Study

- ❑ SPADs are the primary detector candidate for the OGS
- ❑ Payload lasers will be driven by a TBD laser driver and FPGA

Feature \ Modulator	<u>LiNbO3</u>	<u>Silicon Photonic Transmitter</u>	<u>III-V Electro-Optic Polarization Modulator</u>
Phase Modulation	Continuous	Somewhat discrete or nonlinear phase control; requires careful calibration	Limited
Amplitude Modulation	Needs interferometer	Needs interferometer	Limited
Speed	GHz+ precise pulses	Tens of GHz+	Hundreds of MHz
Polarization	Stable	Moderately stable	Moderately stable
Optical Loss	Low	Moderate	Low
Applications	QKD, telecom, phase & amplitude encoding	On-chip BB84, portable QKD	Polarization-encoded BB84 experiments
Benefits	High fidelity, mature, low loss	Compact, integrable, CMOS compatible	Direct polarization control, smaller footprint
Drawbacks	High drive voltage, temperature sensitivity, and cost	Weaker modulation efficiency; needs longer device/higher voltage	Less mature; integration challenges

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Selected in preliminary trade study.

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Mach-Zehnder interferometer

Preliminary Phase Modulator Trade Study

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Drawbacks	High drive voltage, temperature sensitivity, and cost	Weaker modulation efficiency; needs longer device/higher voltage	Less mature; integration challenges

Low Quantum Bit Error Rates (QBER)

$$= \frac{(\text{\# of incorrect bits received})}{(\text{\# of bits sent})}$$

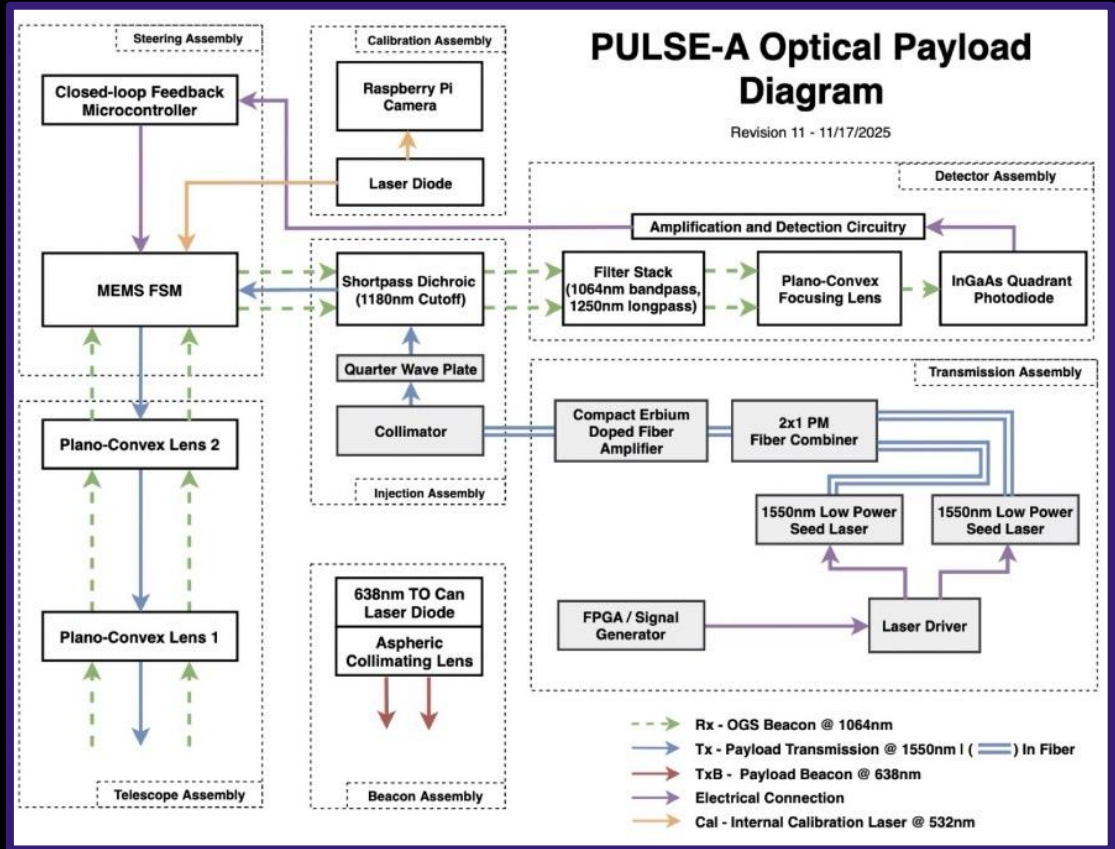
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Applications	QKD, telecom, phase & amplitude encoding	On-chip	BB84
Benefits	High fidelity, mature, low loss	Compact	Control,
Drawbacks	High drive voltage, temperature sensitivity, and cost	Weaker needs voltage	on

Can be accommodated by:
waveguide integration,
optimized RF driver design,
polarization-maintaining fibers,
and thermal stabilization.

Repurposing PULSE-A's Payload Architecture

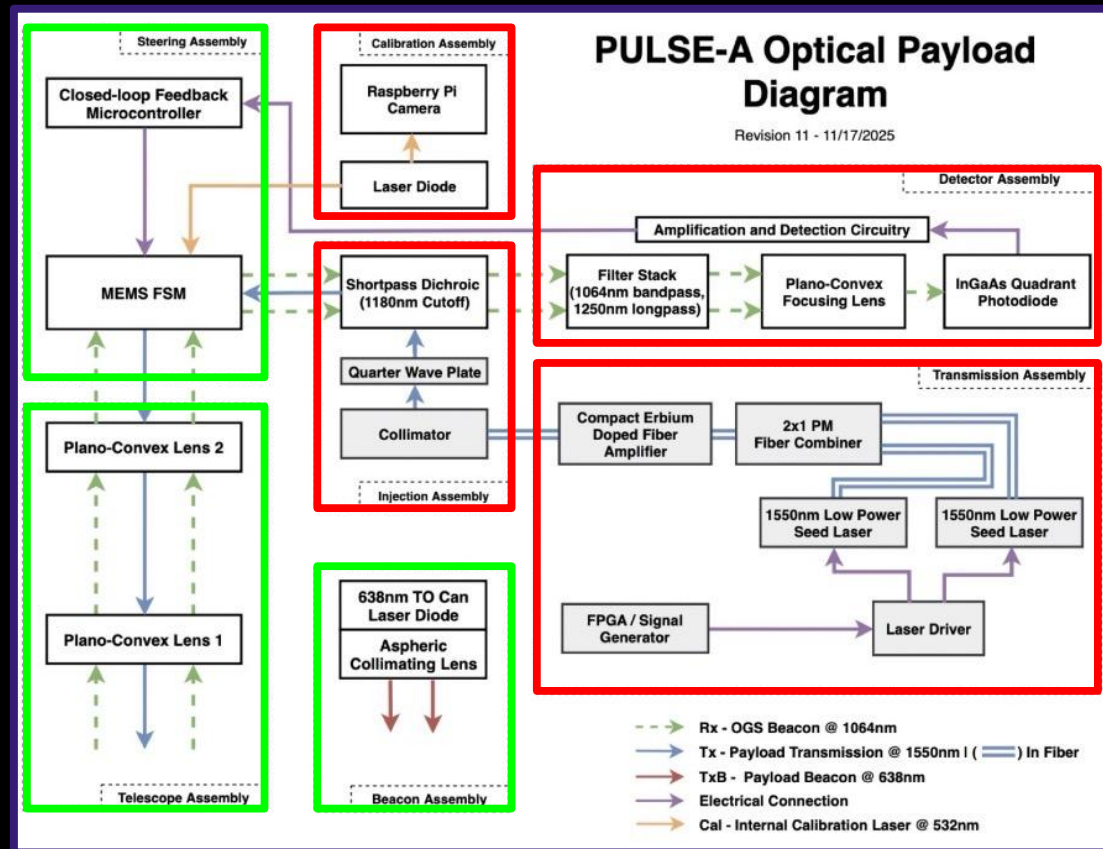


Repurposing PULSE-A's Payload Architecture

Annotation Key:

Assembly likely to remain unaltered or very similar in the PULSE-Q payload

Assembly must undergo significant alterations for the PULSE-Q payload to accommodate the mission objectives



Link Analysis

Transmission wavelength [nm]	1550
Transmit power [dBm]	6.00
Equivalent transmit power [W]	3.9E-3
Received SNR [dB]	7.27
Received power [dBm]	-76.26
Equivalent received power [W]	2.366E-11

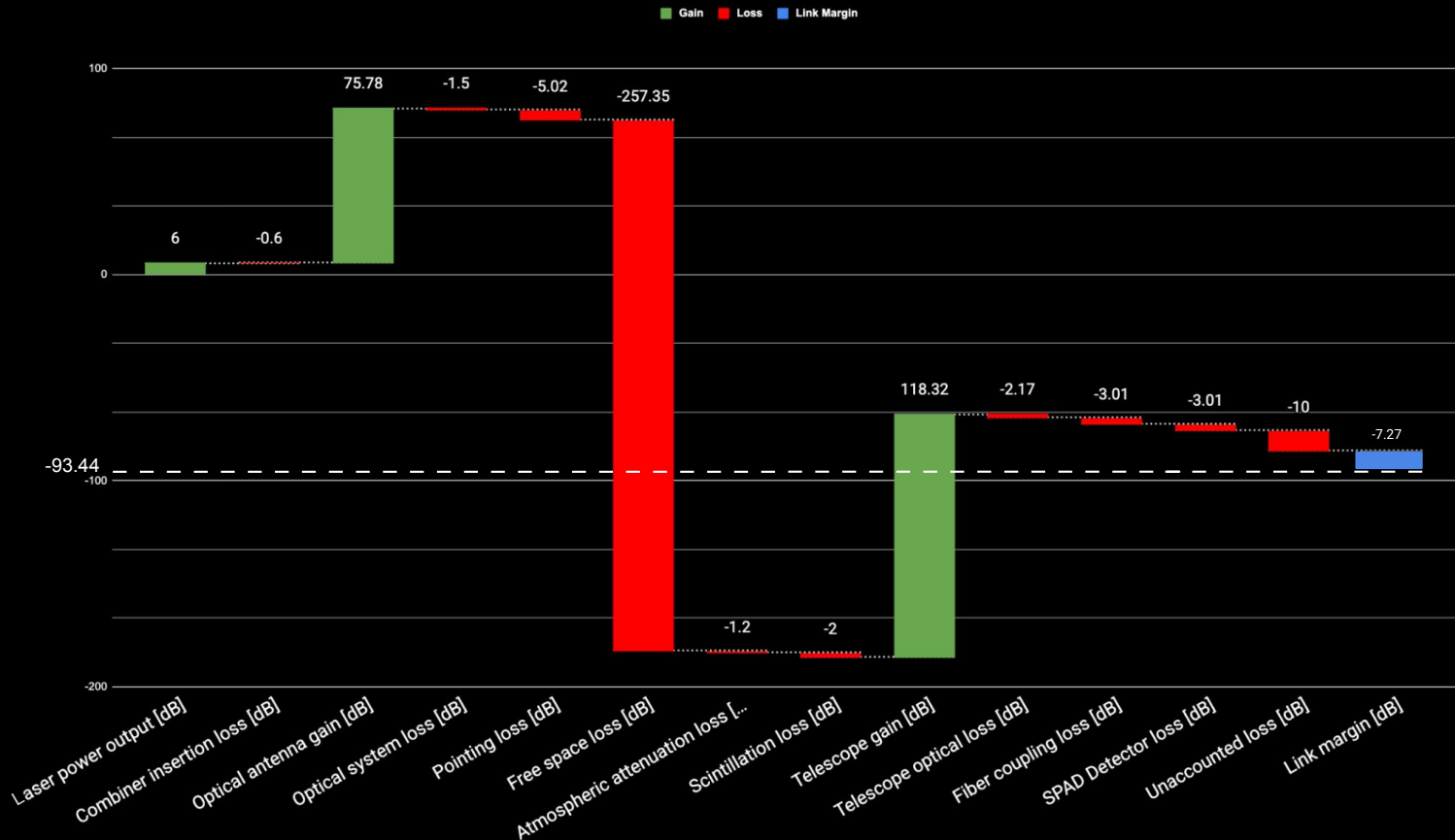
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Received power [dBm]	-76.26
Equivalent received power [W]	2.366E-11

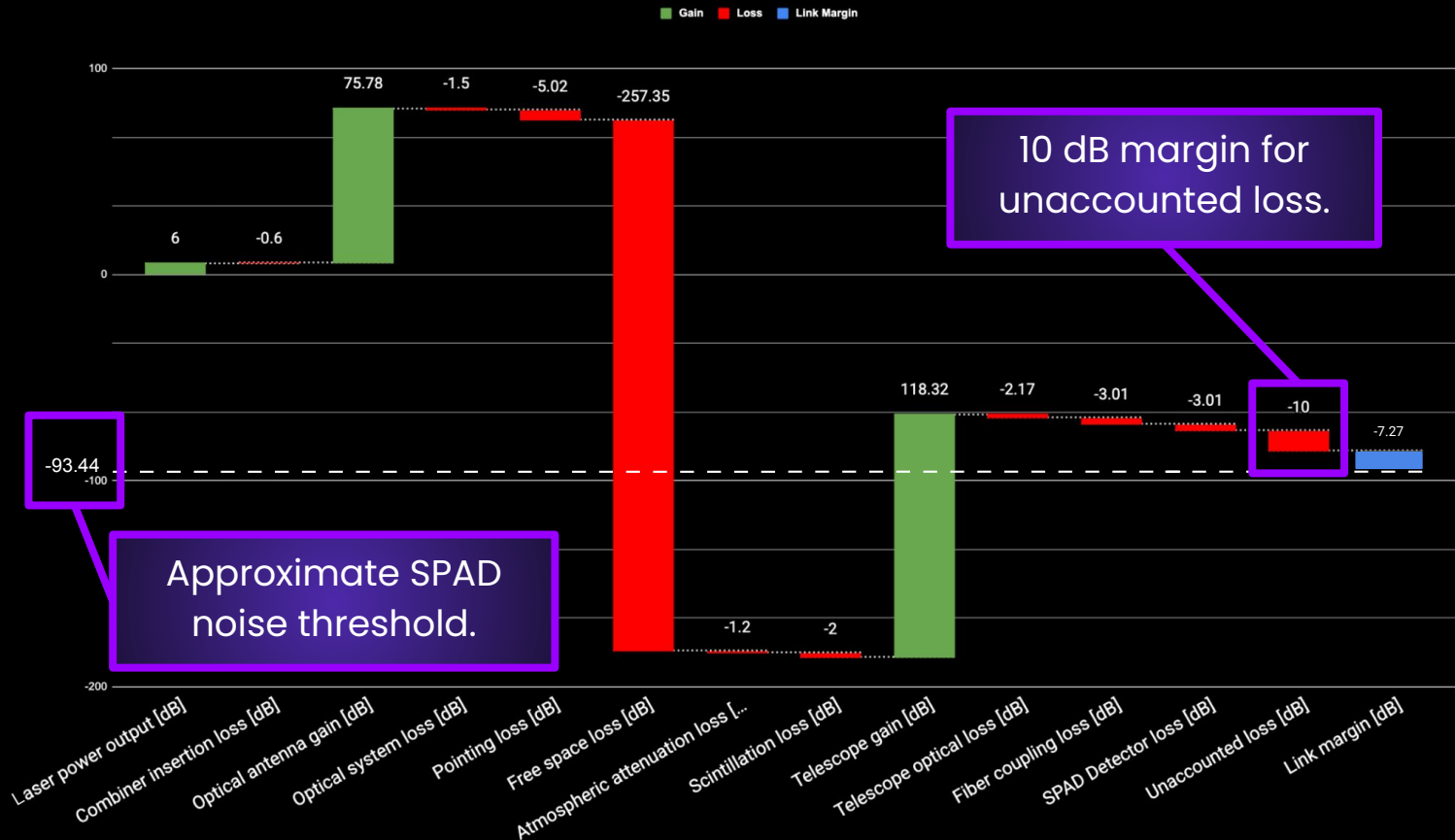
Conservative estimate based on commonly available laser diodes.

SNR>3 dB required in worst case dark count rate, 900 km free space loss, and significant pointing and atmospheric loss.

Link Analysis

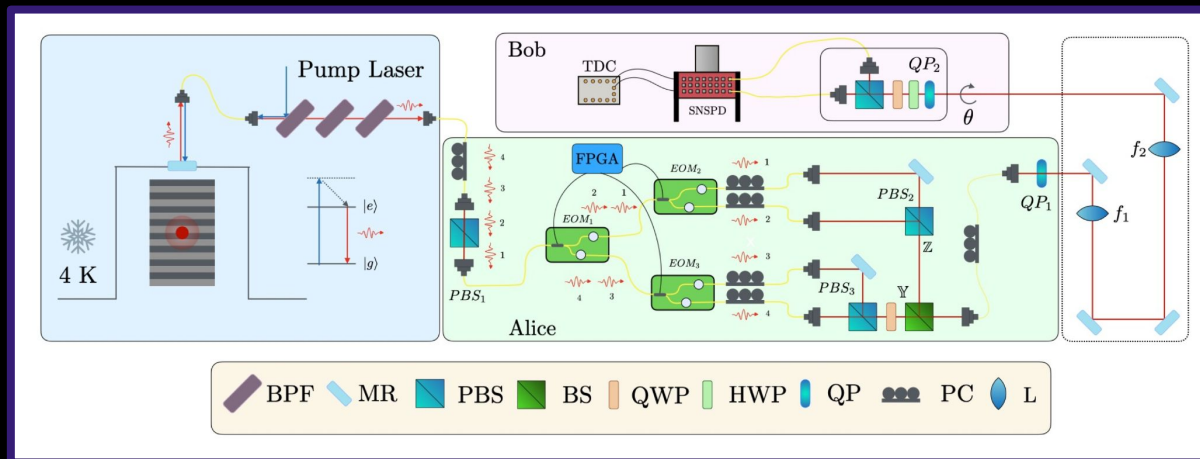


Link Analysis



Alternative Single-Photon Methodologies

- ❑ Choose orthonormal basis for H and V
- ❑ Let the second basis be the superposition basis for D and A
- ❑ Define a computational basis in hybrid polarization/angular momentum states
- ❑ Hybrid communication has been ground-to-ground tested such that
 - ❑ Polarization-orbit hybrid states are rotationally invariant
 - ❑ The system is scalable in the amount of degrees of freedom



Source: Barigelli et al. (2025)

Alternative Single-Photon Methodologies

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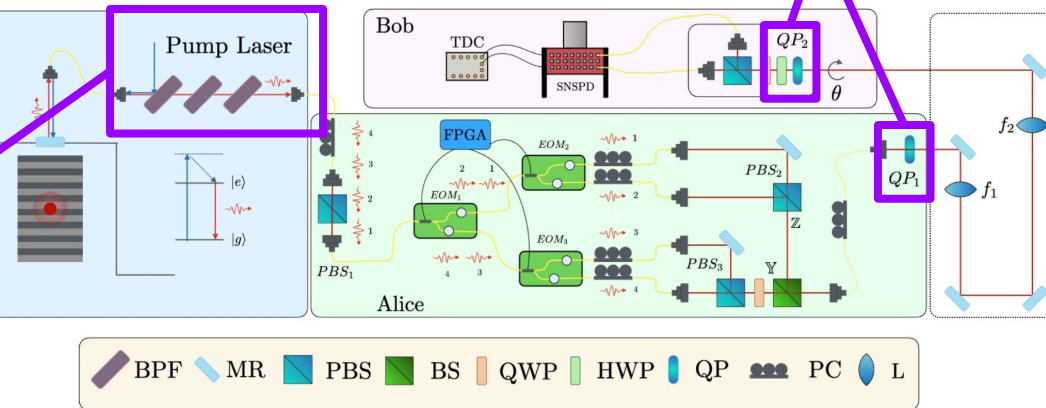
- Hybrid communication has been ground-to-ground tested such that

Q-plates added to convert polarization states to hybrid states and vice versa.

orbit hybrid states are invariant

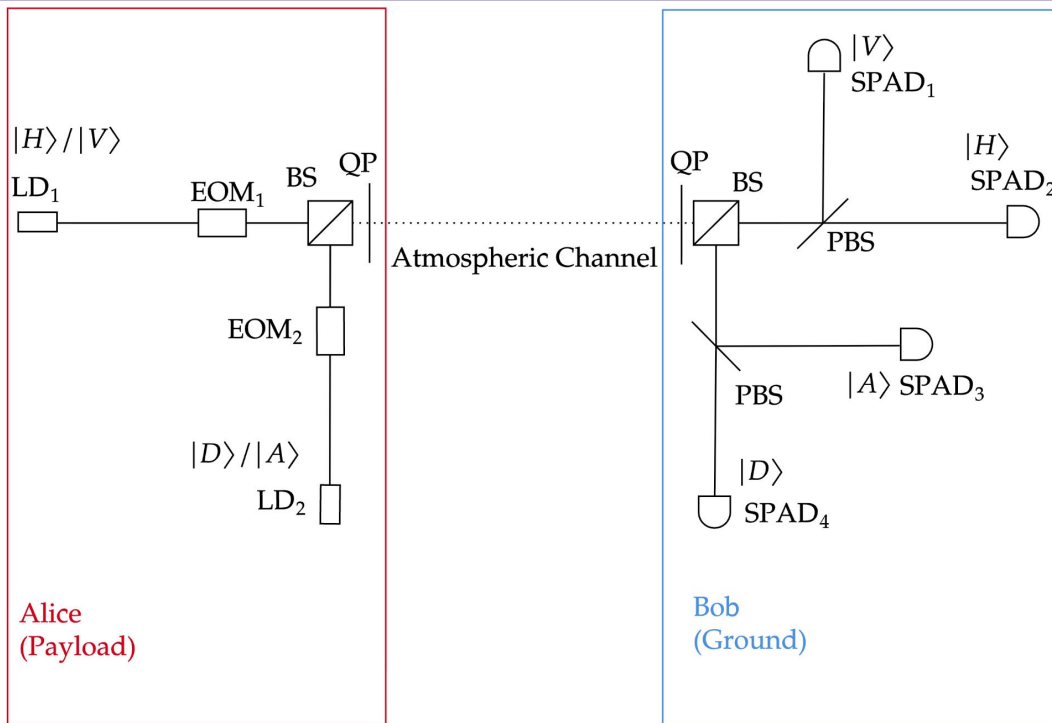
scalable in the degrees of freedom

Single laser system used with electro-optic modulators (costly architecture).



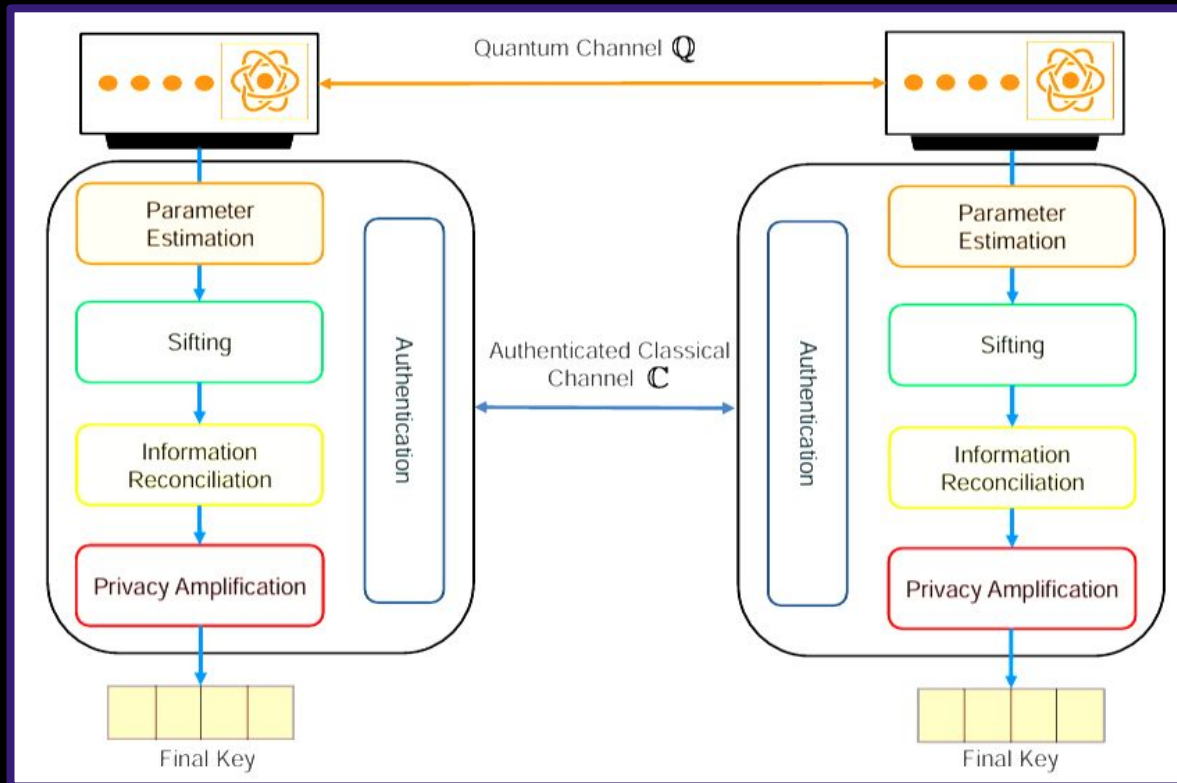
Source: Barigelli et al. (2025)

PULSE-Q Optical System Consideration



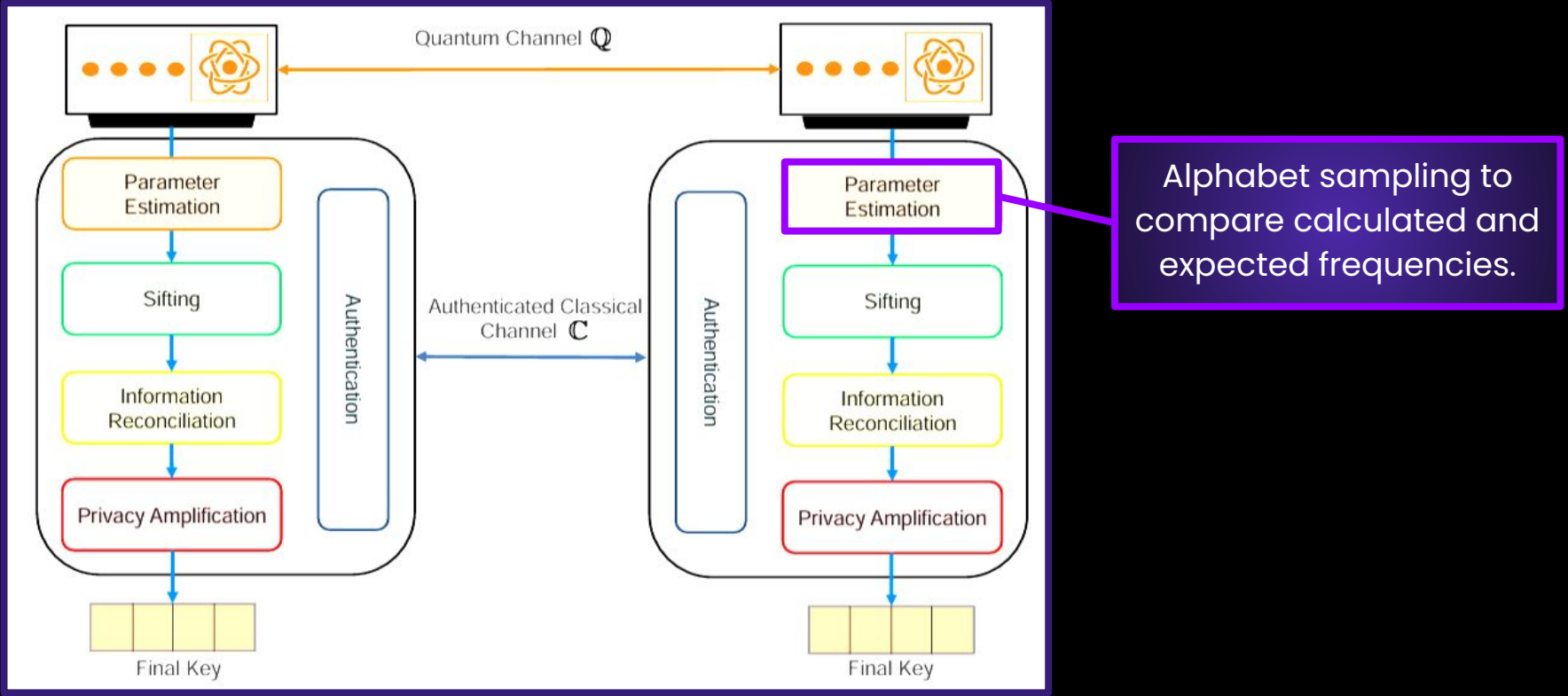
- ❑ Possible alternative to quad-laser setup
- ❑ BB84 protocol with 2 lasers
 - ❑ 2 EOMs for payload
 - ❑ Q-plate for invariant transmission (tentative)
- ❑ For SWaP concerns, this system may be preferable

Data Processing Methodology



Source: Yi Luo et al. (2024)

Data Processing Methodology

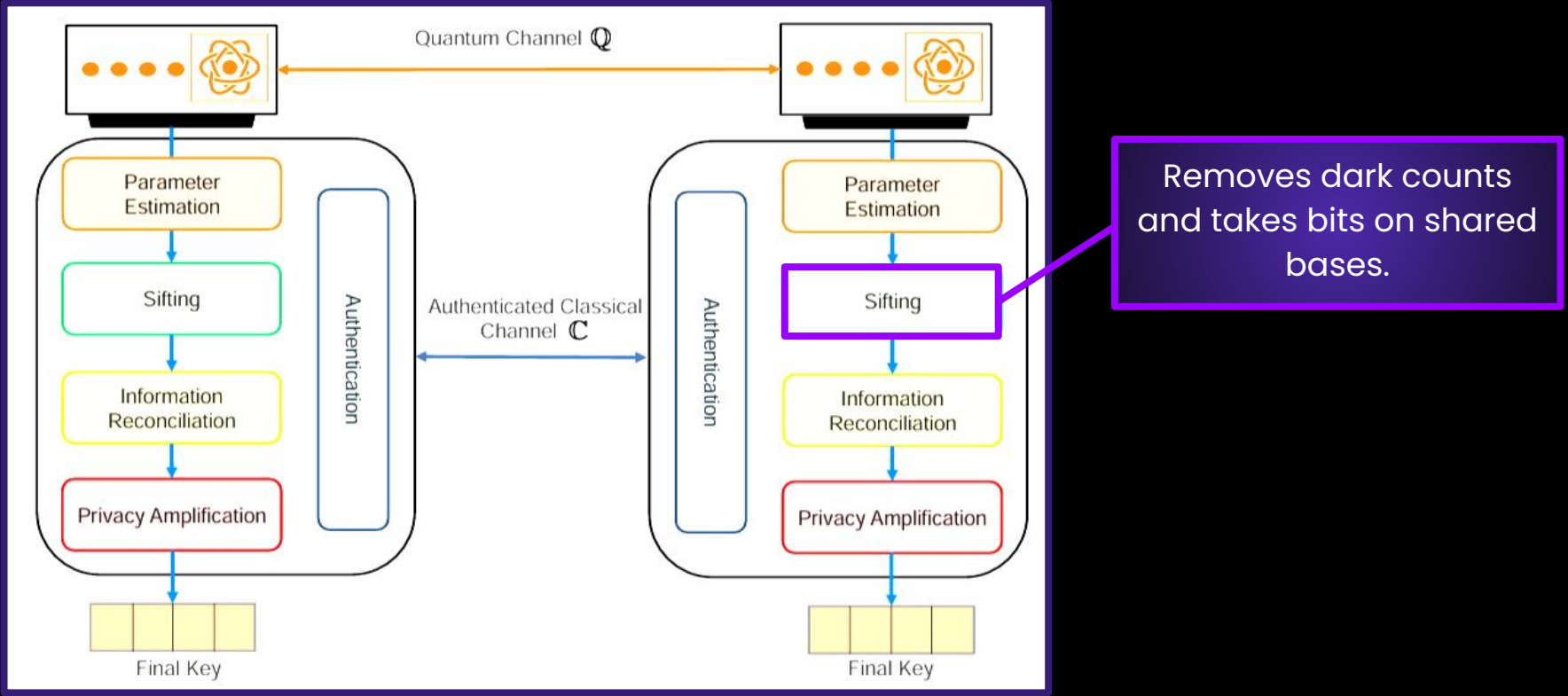


Source: Yi Luo et al. (2024)

Parameter Estimation

- ❑ Alphabet for BB84 consists of: $(a, b)_X$
 - ❑ Their shared basis $X \in \{X, Z\}$
 - ❑ Alice's and Bob's measured bits $a, b \in \{0, 1\}$ in the shared basis
- ❑ Parameters to Estimate: Block size, key rate, security level, abort threshold, yield rates, error rates
 - ❑ Crucial to calculating final secure key rate
 - ❑ Aborts protocol if estimated ER is too high or yield is too low
- ❑ In case of decoy-state implementation, Clopper Pearson (CP) method minimizes sacrificed bits for conservative error confidence intervals
 - ❑ Accounts for finite-key length differences in yield and error rates to asymptotic value
 - ❑ Reducing bits used for PE improves final secure key rate
 - ❑ Beta distribution shaped by Total Signal Count and Yield Count estimated in PE

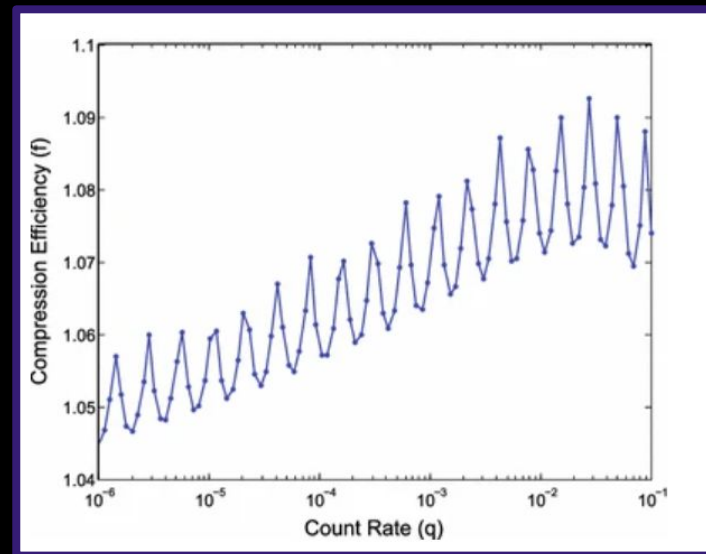
Data Processing Methodology



Source: Yi Luo et al. (2024)

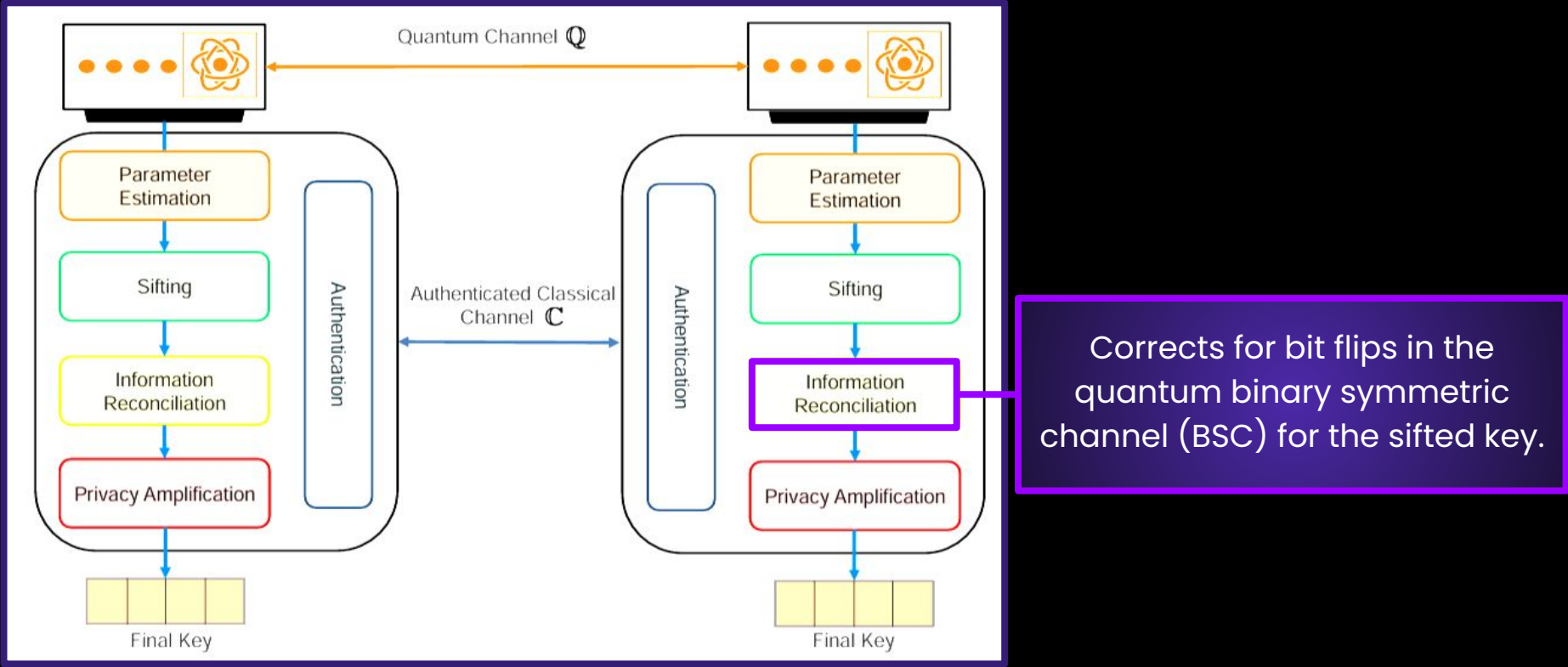
Sifting

- ❑ Optimization for sifting is to maximize likelihood of common basis choice between Alice and Bob through asymmetric basis choice (e.g. BB84 INFO-Z)
 - ❑ $p_z > 0.5 \mid p_z + p_x = 1$ where info is shared only on Z basis
 - ❑ p_z should be optimized to account for minimizing Eve's shared information
- ❑ In practical settings, sifting is a high-latency step. Modified Zero Run Length (MZRL) Encoding provides compression to address this bottleneck
 - ❑ Preserves $O(m)$ time complexity for raw key length m
 - ❑ Increases space complexity to exponential



Source: Qiong Li et al. (2015), Fig. 7

Data Processing Methodology



Source: Yi Luo et al. (2024)

Information Reconciliation

Error Estimation:

- ❑ Communicates sacrificed bits to provide estimates for QBER
 - ❑ QBER estimates allow for optimal code structures and channel modeling
- ❑ For similar reasoning to PE, producing precise error estimates for low key cost is an area of important research.
 - ❑ Regular LDPC code syndrome comparison provide maximum likelihood error estimates

Error Correction:

- ❑ Applies hard input decoders to analog SPAD measurements
 - ❑ SISO decoders are more accurate, though they require robust channel models to estimate bit LLRs
- ❑ Important Heuristics to Consider: QBER, Reconciliation Efficiency, Code Rate, FER, Throughput
- ❑ Primary Decoding Algorithms: Cascade, Winnow, Turbo, LDPC, Polar

Error Correction

Low Density Parity Check (LDPC) Codes:

- ❑ Use Tanner graph belief propagation (BP) to determine bit values based on Alice's LDPC checksum syndrome
- ❑ Pros: Parallelizable, optimized for low complexity, low latency, more popular
- ❑ Cons: Generically high memory, minimum error floor
- ❑ Optimizations: Irregular codes, blind codes, adaptive computational loading, a priori QBER-based codes, belief lookup

Polar Codes:

- ❑ Synthetic channel polarization through recursive XOR structuring
- ❑ Pros: Channel capacity achieving, low latency, higher reconciliation efficiency
- ❑ Cons: Large key lengths to exhibit polarization, recursive, higher complexity (channel Monte Carlo simulations)
- ❑ Optimizations: Successive cancellation lists

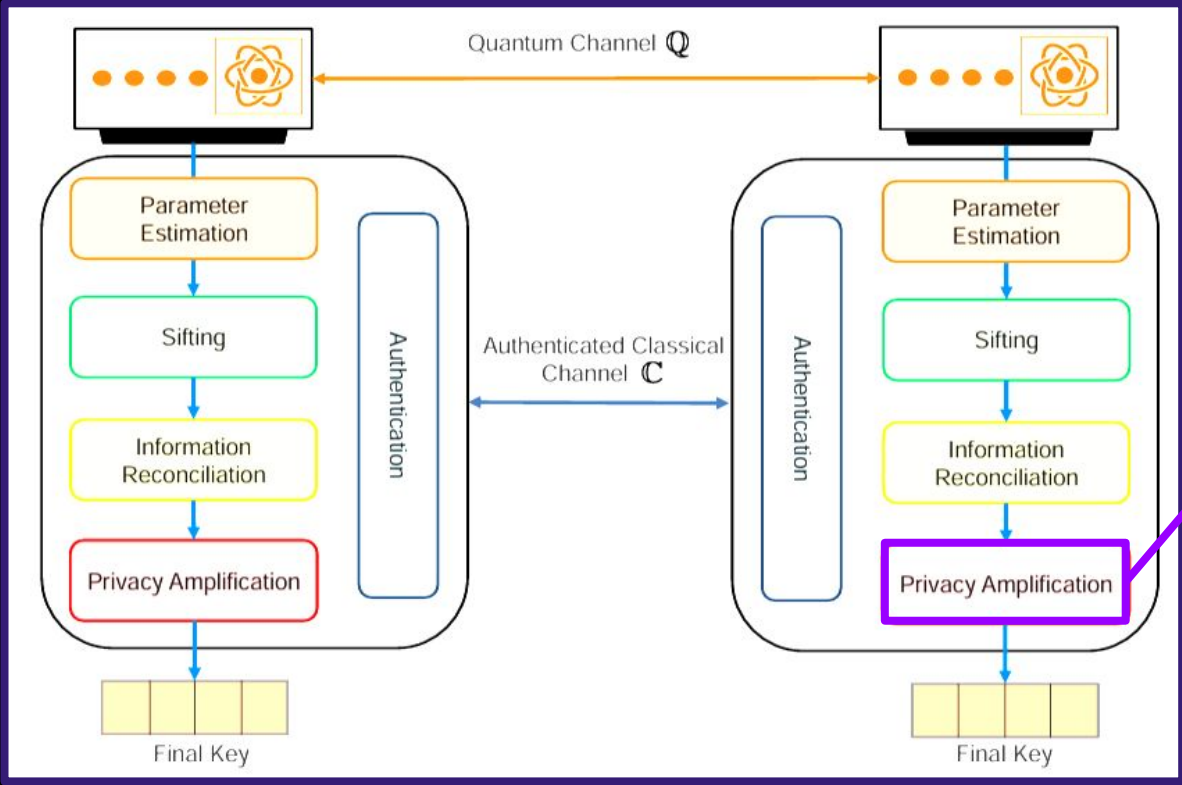
Error Correction

TABLE V: Comparison between turbo, LDPC and polar decoders included in this paper.

Parameter	Turbo decoder	LDPC Decoder	Polar Decoder
Computational complexity	Higher for most coding rates. Low at low coding rates	Lower for most coding rates. High at low coding rates	Lower for most coding rates
Information throughput	Low. Enhanced by fully-parallel structure	High for high coding rates and parallel structure	High for pipeline structure
Area efficiency	Low for most coding rate. High at low coding rates.	High for most coding rates. Low at low coding rates.	High for most coding rates.
Energy efficiency	Low for most coding rates. High at low coding rates.	High for most coding rate. Low at low coding rates.	High for most coding rates.
Error correction performance	Similar at long block lengths.	Similar at long block lengths.	Superior for short block lengths.
Flexibility	High flexibility	Partial flexibility	Very high potential

Source: Shuai Shao et al. (2019)

Data Processing Methodology

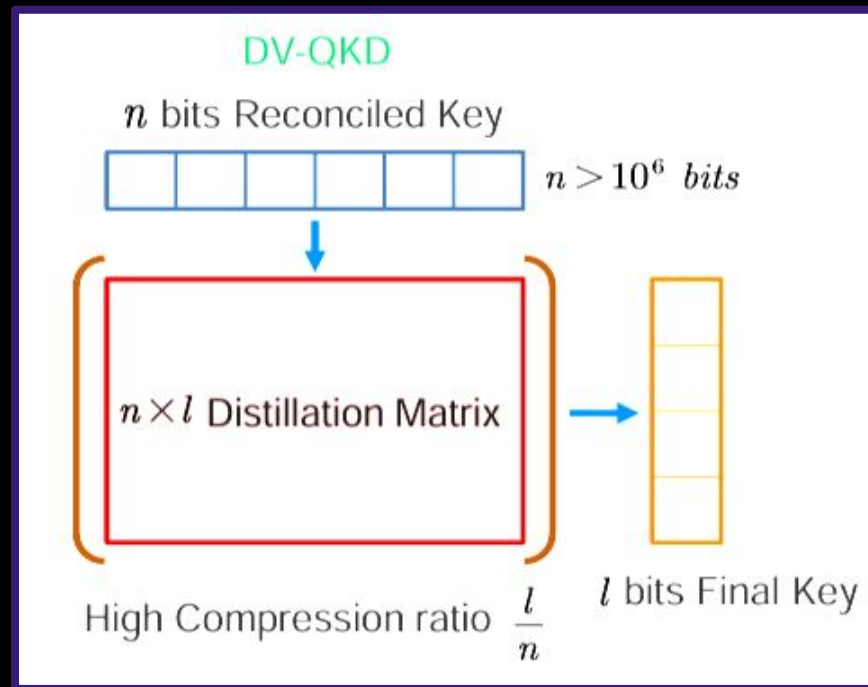


Partial information exposure via classical channel is addressed through a final key mapping.

Source: Yi Luo et al. (2024)

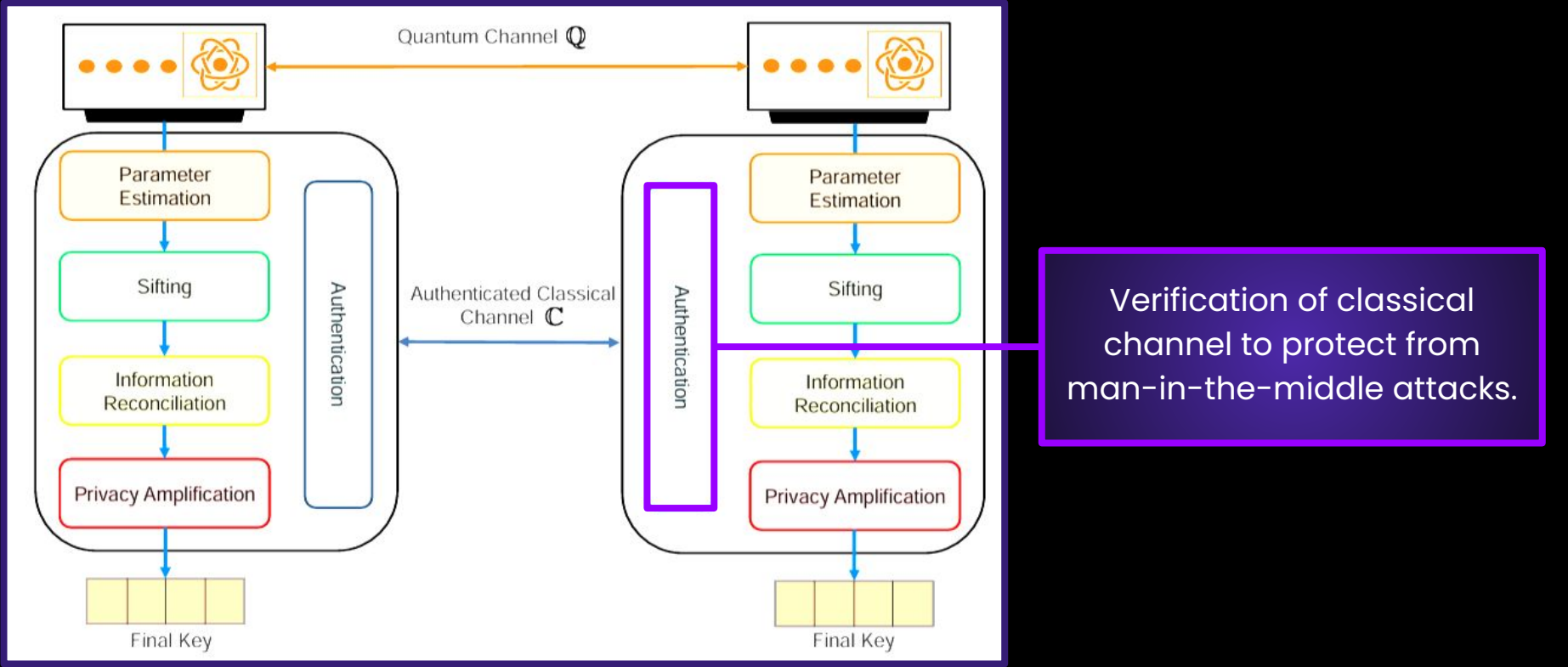
Privacy Amplification

- ❑ Select from a family of universal-2 hash functions to minimize final key collision between Eve's key and the corrected key
- ❑ Hash functions can apply a large memory burden
 - ❑ Easier compression using a class of hash functions that are Toeplitz Matrix



Source: Yi Luo et al. (2024)

Data Processing Methodology



Source: Yi Luo et al. (2024)

Channel Authentication

- ❑ Channel authentication runs parallel to the rest of the post-processing; it is not a throughput bottleneck
- ❑ Different models that primarily use hashing to verify the key is being successfully produced
 - ❑ Certain models sacrifice a portion of the key for authentication, drastically affecting final key rate

Simple Model

- ❑ R_p = source pulse rate (Hz)
- ❑ η = total link / detector efficiency
(derived using link loss, fraction of photons reaching receiver)
- ❑ f_{sift} = sift factor (~ 0.5 for two bases)
 N_{sift}, N_c = bits after basis reconciliation, classical bits for error correction
- ❑ f_{EC} = error correction overhead per sifted bit (for QBER of 5% we can use concept of entropy to get ~ 0.3)

$$N_{\text{det}} = R_p \cdot t_{\text{pass}} \cdot \eta$$

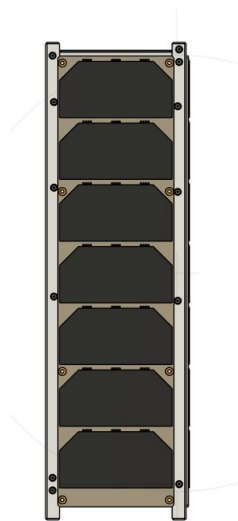
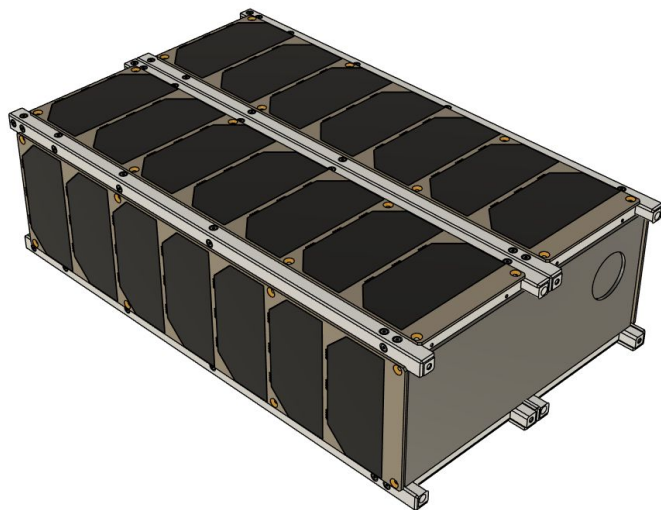
$$N_{\text{sift}} = f_{\text{sift}} N_{\text{det}}$$

$$N_c = N_{\text{sift}} (1 + f_{\text{EC}})$$

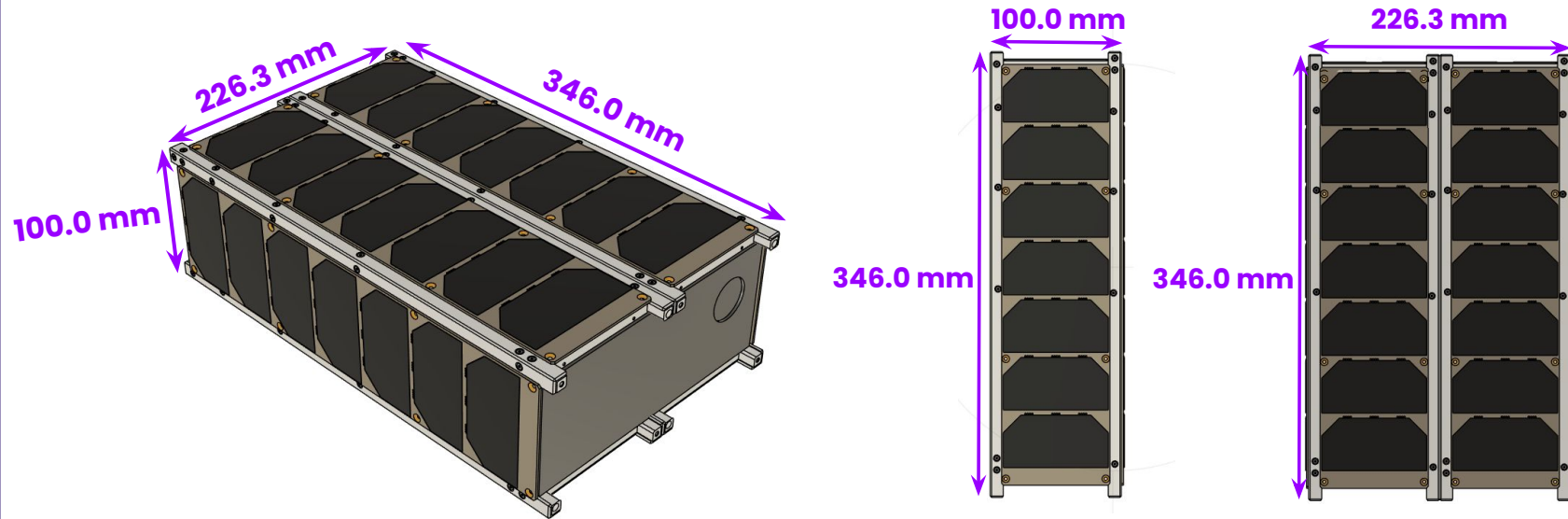
$$R_c = \frac{N_c}{t_{\text{pass}}}$$

Spacecraft System Concept

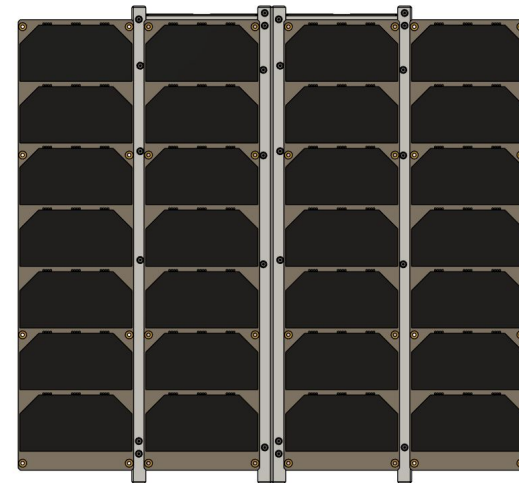
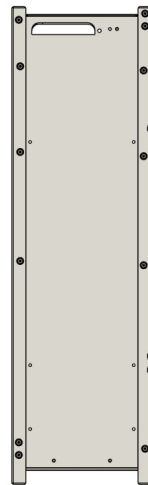
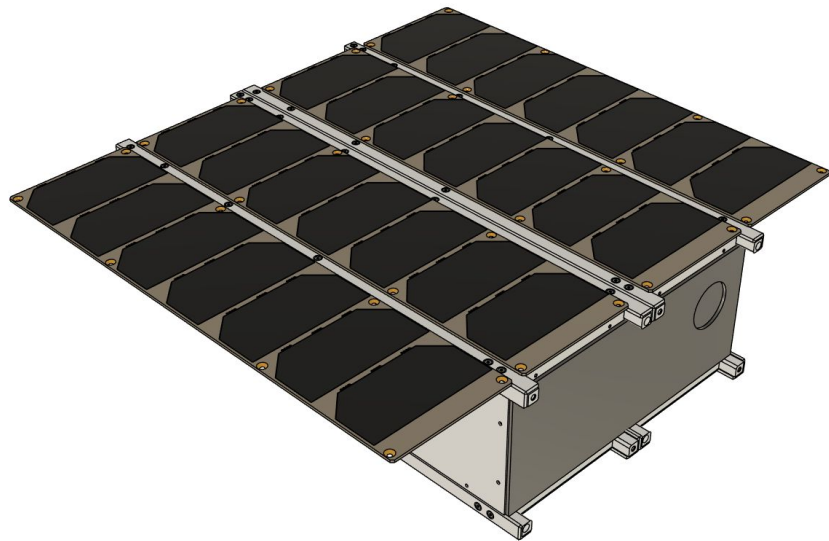
Preliminary Exterior Configuration: Stowed / No Deployable Solar Panels



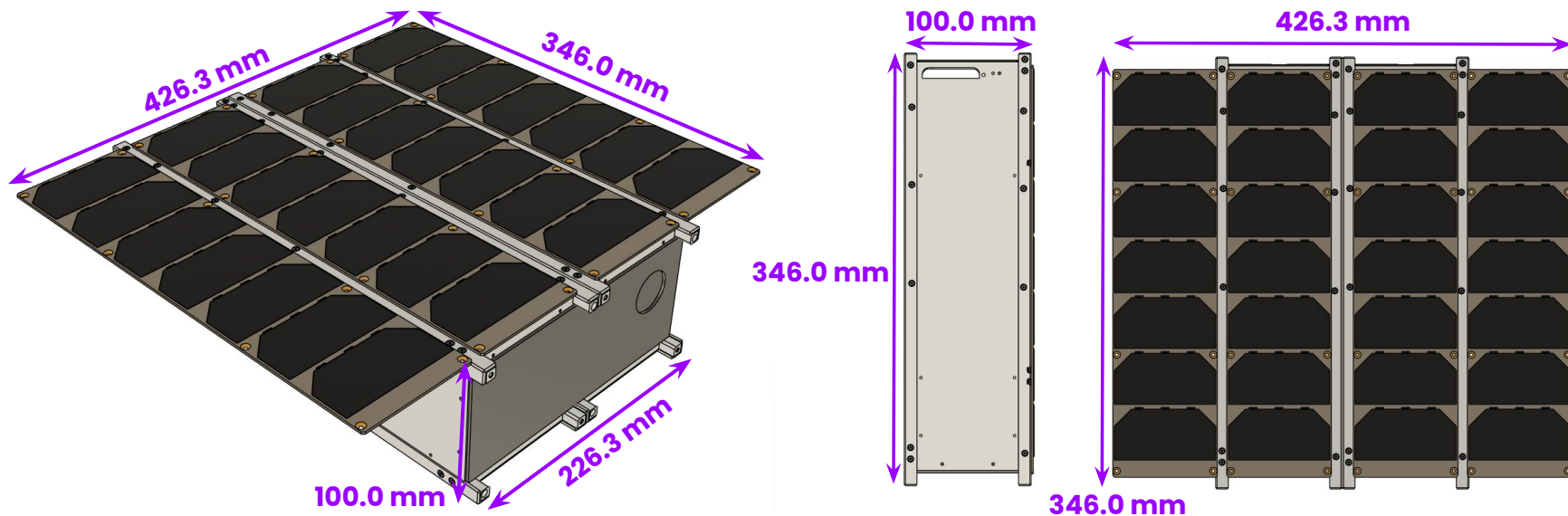
Preliminary Exterior Configuration: Stowed / No Deployable Solar Panels



Preliminary Exterior Configuration: Deployable Solar Panels (Configuration Candidate)



Preliminary Exterior Configuration: Deployable Solar Panels (Configuration Candidate)



Mass Budget: No Deployable Solar Panels

Subsystem	Mass (g)
Optical Payload	2,000
ADCS	1,000
Electrical Power	2,800
Command & Data Handling	300
RF Communications	150
Structure	1,600
Thermal	250
Total	8,100
Margin (Maximum Allowable Mass = 12 kg)	3,900 (32.5%)

Mass Budget: With Deployable Solar Panels

Subsystem	Mass (g)
Optical Payload	2,000
ADCS	1,000
Electrical Power	3,500
Command & Data Handling	300
RF Communications	150
Structure	1,600
Thermal	250
Total	8,800
Margin (Maximum Allowable Mass = 12 kg)	3,200 (26.7%)

Volume Budget

Subsystem		Volume (cm ³)	
Optical Payload		1,900	
ADCS		850	
Electrical Power		425	
Command & Data Handling		200	
RF Communications		125	
Additional Bus Hardware		300	
Total		3,800	
Margin	(Frame Volume ≈ 5,700 cm ³)	1,900	(33.3%)

Power Budget

	Approximate Power Draw	Notes
Payload Operations Power Draw	13 W	Average power draw from various papers (all 9-16W)
Bus Power Draw	7 W	Average power draw with no payload operations active
Energy Consumed During Eclipse w/ Optical Downlink	5.9 Wh	Calculated from a 45-minute eclipse and 3-minute payload operations
Battery Capacity Required	19.7 Wh	Calculated with a depth of discharge of 30%
Battery Capacity	74 Wh	Pumpkin Space Systems Battery Module 2 capacity at 2500 mAh
Solar Panel Generation (6U Area)	17.7 Wh	Power generated by 6U/12U area SpaceManic RA Solar Panels (used by PULSE-A) at EOL efficiency 25.7%
Solar Panel Generation (12U Area)	35.47 Wh	

RF Communications

- ❑ RF links carry auxiliary data, telemetry, and public (not QKD-protected) information in support of mission operations
- ❑ QKD protocols require multiple authenticated classical exchanges within limited pass durations (~8 min)
- ❑ UHF/VHF: Comparably low throughput (~10 kbps); limited mission support capabilities outside of TT&C
- ❑ S-band: Moderate throughput (~2–20 Mbps); supports basis reconciliation, error correction, privacy amplification
- ❑ X-band: High throughput (~10–100 Mbps); supports large reconciliation blocks or bulk science payloads; highly costly

Communication Band	Data Rate	8 min. Capacity	Typical Role in QKD
VHF/UHF	10 kbps	0.6 MB	TT&C only
S-band	2 Mbps	120 MB	Classical QKD metadata
X-band	50 Mbps	3 GB	Bulk classical data / payload

RF Communications

S-band or X-band communications are more suitable for QKD mission operations. However, selecting a communication band other than VHF/UHF requires significant upgrades to PULSE-A’s RFGS for use by PULSE-Q.

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Risk Assessment & Mitigation

Major Mission Risks

ID	Potential Risk	Description
A	Software Failure	Radiation corrupts software, bugs
B	Quantum/Optical Loss	Photon/information loss, inability to perform QKD
C	Battery Failure	Batteries become inoperative, degrade faster than expected
D	Radiation Damage	Radiation damages components
E	Thermal	Components damaged by heat
F	Launch Vehicle Failure	Destruction of launch vehicle, incorrect orbit
G	Student Turnover	Students graduate or leave the development team

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Likely reparable during mission operations unless effects are extreme.

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Main risk differing from PULSE-A. Procedural caution and in-depth validation necessary for QKD equipment.

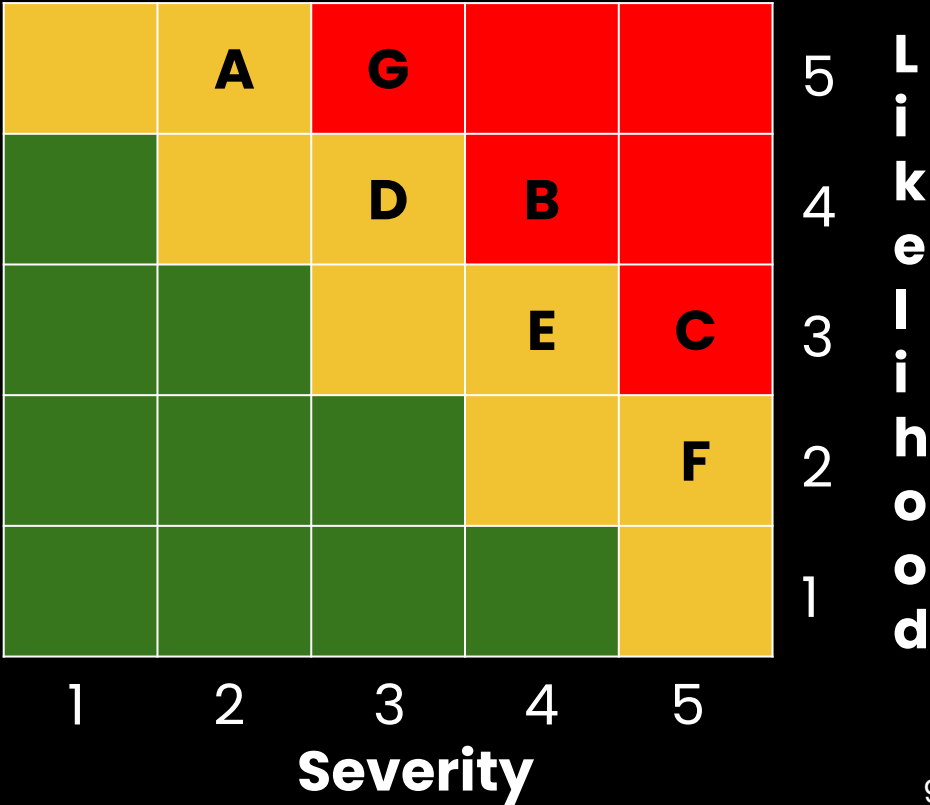
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Graduation and natural attrition necessitate planning training for role transitions and developing skills to continue the mission without resulting delays or technical challenges.

Pre-Mitigation Risk Matrix

ID	Potential Risk
A	Software Failure
B	Quantum/Optical Loss
C	Battery Failure
D	Radiation Damage
E	Thermal
F	Launch Vehicle failure
G	Student Turnover



Projected Mitigation Strategies

ID	Potential Risk	Mitigation
A	Software Failure	Shield sensitive components, redundancy
B	Quantum/Optical Loss	Testing, hardware, entanglement
C	Battery Failure	COTS components, battery shielding, redundancy
D	Radiation Damage	Shielding, testing
E	Thermal	Thermal shielding, testing
F	Launch Vehicle Failure	Select low-risk launch provider
G	Student Turnover	Consistent recruitment, thorough documentation, training

Risk B requires extra attention, as quantum optics are PULSE-Q's primary novelty over PULSE-A. Rigorous testing and ground twin QKD operations will be conducted to ensure low risk likelihood.

Projected Mitigation Strategies

ID	Potential Risk	Mitigation
A	Software Failure	Shield sensitive components, redundancy
B	Quantum/Optical Loss	Testing, hardware, entanglement
C	Battery Failure	COTS components, battery shielding, redundancy
D	Radiation Damage	Shielding, testing
E	Thermal	Thermal shielding, testing
F	Launch Vehicle Failure	Select low-risk launch provider
G	Student Turnover	Consistent recruitment, thorough documentation, training

Risks C and F are the most severe—effects have significant consequences for the overall system. Careful selection of low-risk commercial vendors/partners reduce risk likelihood.

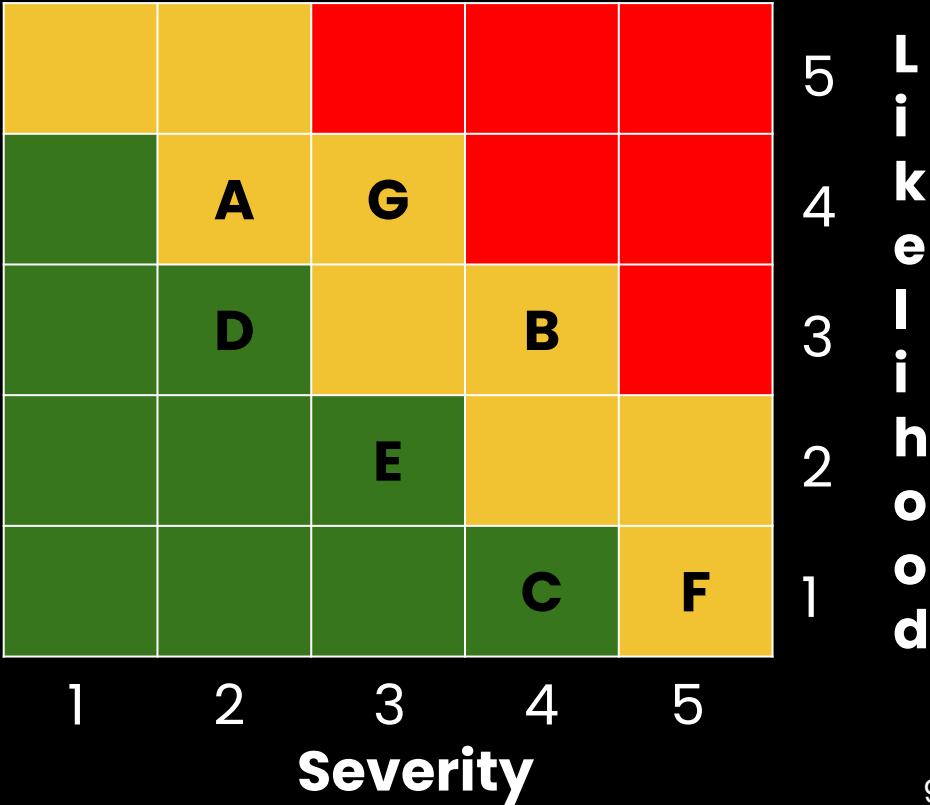
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B	Quantum/Optical Loss	Testing, hardware, entanglement
C	Battery Failure	COTS components, battery shielding, redundancy
D	Radiation Damage	Shielding, testing
E	Thermal	Thermal shielding, testing
F	Launch Vehicle Failure	Select low-risk launch provider
G	Student Turnover	Consistent recruitment, thorough documentation, training

Risks D and E have the potential to impact all subsystems to varying extents. Shielding, testing, and simulations will reduce likelihood and severity of the risks' consequences.

Post-Mitigation Risk Matrix

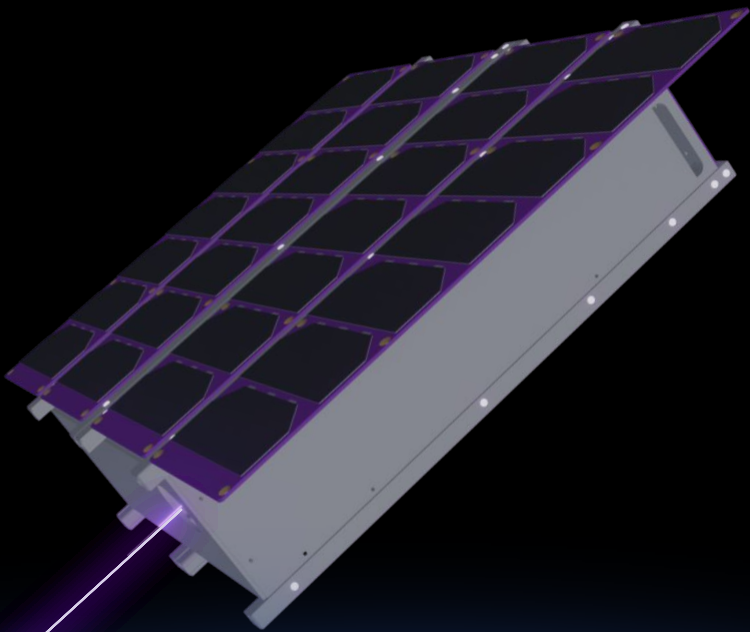
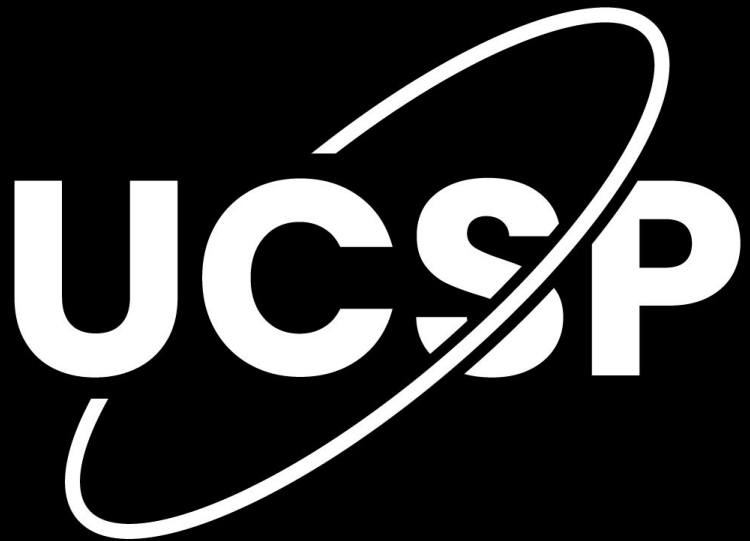
ID	Potential Risk
A	Software Failure
B	Quantum/Optical Loss
C	Battery Failure
D	Radiation Damage
E	Thermal
F	Launch Vehicle failure
G	Student Turnover



Conclusion

MCR Exit Criteria

- ❑ The principal mission objective and guiding priorities in mission development are agreed upon and finalized
- ❑ The mission's merit is clearly stated and justified
- ❑ The operational concept is demonstrated to be feasible and achieve top-level mission objectives if successfully implemented
- ❑ Technical margins are adequate to demonstrate mission feasibility with respect to cost, schedule, spacecraft mass, spacecraft volume, and communications links
- ❑ The primary candidates for system architecture are identified, and resulting key areas of focus are selected to guide engineering in Phase A (Concept Development)
- ❑ The most significant technical and programmatic risks are identified; systems engineering strategies are outlined to manage risk within financial and schedule constraints



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Pulse-A Link Budget